

SEED STENCILS: HIGH-ORDER FINITE DIFFERENCES AND RBF-FD THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

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ABSTRACT. Between a material whose parameters vary smoothly on the scale of the grid and one with a jump lies the case of an edge that is smooth but changes over a distance δ far smaller than the grid spacing h . Standard high-order finite differences sample the coefficients at the nodes and lose their order there until $h \lesssim \delta$, and the interface-aware stencils built for a jump do not apply, since there is no jump. We build stencils on the same equispaced (1-D) or scattered (2-D) nodes whose basis functions are not monomials but *seeds*: the $t = 0$ profiles of solutions of the wave equation that are polynomial in time, continued through the edge by ordinary differential equations in the coordinate normal to it. The seeds form an extended complete Chebyshev system for every δ , so the stencil solve is nonsingular, and as $\delta \rightarrow 0$ they reduce to the translated polynomial basis of the interface-aware stencils, so one construction covers the jump, the sub-grid edge and the resolved edge. In one dimension, standard fourth-order finite differences have a knee at $h = \delta$ while the seed stencils are fourth order at every resolution with the errors of the jump case. In two dimensions, for the elastic wave equation discretised with radial basis function-generated finite differences (RBF-FD) on scattered nodes, the seeds of a straight edge reduce to triangular systems of ordinary differential equations in the normal coordinate; the seed-augmented stencils are stable at the standard hyperviscosity, cut the error through an unresolved edge by up to an order of magnitude or more on the node sets tested, and lead to the rule: seed when $\delta \leq h$. The same seeds marched along the true normal handle a curved edge at the same accuracy on those node sets.

1. INTRODUCTION

High-order finite differences for wave propagation are in good shape at both ends of the range of media. When the material parameters vary smoothly on the scale of the grid, sampling them at the nodes costs nothing and a fourth-order stencil stays fourth order. When they jump, the stencils that straddle the jump can be rebuilt from piecewise polynomials that satisfy the interface conditions, and the scheme keeps its order on a grid that takes no notice of the interface [Mar16, MF17a]. Between the two lies a case that is neither. An edge can be smooth, so that no interface condition applies, and still change over a distance δ smaller than the grid spacing

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h . Any smooth model contains such edges once it is discretised coarsely enough, and any jump becomes one under the slightest smoothing.

A standard stencil through such an edge samples the coefficients at its nodes, sees a jump, and treats it as if the coefficients were smooth in between, which is the first-order failure of the jump case. It recovers only once $h \lesssim \delta$. In the one-dimensional experiment of Section 4, fourth-order finite differences (FD4) through an edge of width $\delta = 0.0025$ converge at rates 1.3 and 2.2 while $h > \delta$, drop by a factor 200 between $n = 400$ and $n = 800$ nodes ($h = 2\delta$ to $h = \delta$), and are fourth order after that: a knee at $h = \delta$. The two existing repairs do not reach this regime. The interface-aware stencils built for a jump are the $\delta \rightarrow 0$ limit of the right construction, as Section 3 shows, but their weights are off at first order in δ/h , by 10% of the largest weight at $\delta = h/10$ and 1% at $\delta = h/100$ in the example of Section 3.2. The other route expands the coefficients in Taylor series about the edge, as the interface-aware construction can for a wave speed that varies smoothly up to an interface [Mar16, MF17a] and as polynomial quasi-Trefftz spaces do in general [IGMS23]; but a profile such as $\tanh(x/\delta)$ has poles at $\pm i\pi\delta/2$, so the expansion converges only within $\pi\delta/2$ of the edge while the stencil reaches $2h \gg \delta$.

The construction of this paper keeps everything about the scheme except the basis functions of the stencils that see the edge. In place of the monomials $1, x, x^2, \dots$ a stencil is made exact on *seeds*: the $t = 0$ profiles of solutions of the wave equation that are polynomial in time. For the operator $L_u = \rho^{-1}\partial_x K \partial_x$ that governs the particle velocity, such a profile satisfies $L_u^m g = 0$ for some m , and with constant coefficients these profiles are exactly the polynomials. Anchored at the stencil's evaluation point x_e they form a chain, $\phi_0 = 1$, $K\phi'_1 = K(x_e)$, $L_u\phi_k = k(k-1)c_e^2\phi_{k-2}$ with $\phi_k(x_e) = \phi'_k(x_e) = 0$, which is integrated through the edge as a first-order system of ordinary differential equations that never differentiates the coefficients: one short march per stencil that sees the edge. Three facts make this a method rather than a trick. The seeds form an extended complete Chebyshev system for every δ , so the per-stencil solve is nonsingular on any distinct nodes, including at $\delta = 0$ where the seeds have kinks, with the conditioning of a Vandermonde matrix (condition numbers 20 to 60 from $\delta = 10^{-5}$ to 1). The truncation error of the seed stencil on a solution is $O(h^4 \partial_x \partial_t^4 u)$, bounded independently of δ , where FD4's $O(h^4 u^{(5)})$ carries $u^{(5)} \sim \delta^{-4}$ inside the edge. And the chain read in the weak sense across a jump is the translated Taylor basis of the interface-aware stencils, so one construction covers the jump (algebra), the sub-grid edge (an ODE march) and the resolved edge, where the seeds and the monomials give different weights, first order in h/δ apart, and both are fourth order on solutions.

The same idea carries to two dimensions, to the elastic wave equation on scattered nodes discretised with radial basis function-generated finite differences (RBF-FD). For a straight edge the material depends on the normal coordinate only, and an ansatz polynomial in the tangential coordinate turns

the chain into triangular systems of one-dimensional ODEs in the normal coordinate: all 30 seeds of one stencil, the monomials to degree four in each velocity component, march as one linear system of 600 states anchored at the evaluation node. The seeds replace the polynomial augmentation of the interface-aware RBF-FD stencils; the Gaussian part of the stencil and the coupled saddle-point solve stay as they are. For a curved edge the same seeds are marched along the true normal through the stencil's foot point, an approximation of zeroth order in the curvature, which is the one the jump stencils already make.

Contributions.

- (1) In one dimension: the seed chain, its nested kernels, its well-posedness as an extended complete Chebyshev system, and the jump construction as its $\delta \rightarrow 0$ limit (Section 3). On the same equispaced grid the seed stencils are fourth order at every resolution, with errors matching the jump case to three digits, while FD4 has its knee at $h = \delta$ (Section 4).
- (2) In two dimensions: the elastic seeds of a straight edge as triangular systems of ODEs in the normal coordinate, marched as one system per stencil; their assembly into seed rows for the elastic operator and hyperviscosity rows that annihilate the seeds; and the finding that the seeded operator is stable at the standard hyperviscosity at every edge width (Sections 5 and 6).
- (3) A rule: *seed when $\delta \leq h$* . Through an edge the nodes never resolve the seeds are fourth order at every resolution tested, $20\times$ below the standard scheme in the velocity and $67\times$ in the spurious tangential velocity at 19,600 nodes, at the seed operator's own resolution floor; the crossover is at $h \approx \delta$; and through a resolved edge the seed stencils are 1.4 to $2\times$ worse than the standard ones, so a resolved edge should not be seeded.
- (4) At oblique incidence the tangential dependence of the seeds is what acts: the full seeds beat the standard scheme by 1.4 to $1.9\times$ and the normal-only ablation is worse than doing nothing through a sharp edge. Every scheme converges at about order 2.5 there, the jump-aware stencils included, because the mode-converted shear waves sit at a pre-asymptotic floor on these node sets; the result is reported as it is.
- (5) Through a curved edge the jump stencils sit at the resolution floor at every node count, and the seeds marched along the true normal equal the flat seeds at their own floor, 2.2 to $7\times$ below the standard scheme at $\delta = 0.005$ and 3 to $12\times$ at $\delta = 0.0025$ on node sets up to 19,600 nodes; the zeroth-order geometry first shows at the finest of them.

- (6) An open implementation in Python with a test for every numerical routine, in which every figure and table of this paper is regenerated by a committed script from a results cache.

1.1. Relation to prior work. Three lines of prior work meet here, and the construction adds nothing to any of them taken alone. Stencils and elements built from local solutions of the operator are the fitted operators of the 1950s and 60s [AS55, Il'69, SG69, RST08] and the exact schemes that followed [Mic93, VSKO20]; the constant-flux seed in particular is the harmonic mean of Tikhonov and Samarskii [TS62, Sam01], the special elements of Babuška and Osborn [BO83, BCO94], the multiscale elements of Hou and Wu [HW97] and the harmonic coordinates of Owhadi and Zhang [OZ07], whose extension to the acoustic wave equation [OZ08] is the closest two-dimensional relative of what follows. Basis functions that are polynomial-in-time solutions of the wave equation are the Trefftz spaces of space-time discontinuous Galerkin methods [KMPS16, BGL17, MP18]; their quasi-Trefftz extension to variable coefficients [IGMS23, IGMPS25, IG25] builds them by Taylor expansion at a point, the route Section 3.4 shows cannot reach an edge thinner than the stencil. For a jump, the interface-aware stencils this work generalises [Mar16, MFSC15, MF17a] stand beside the immersed interface [LL94, ZL97, SMP⁺23], explicit simplified interface [PL01, LP04] and summation-by-parts [MN06, DV14, GW25] treatments, all of which take the jump as given, where standard schemes are first order [Bro84, SV09, VLTR14]. The standing alternative in seismic and electromagnetic finite differences changes the medium rather than the stencil: coefficients averaged over a cell [Bac62, MDEN92, MKV⁺02, KMCK17, JZ24, FRR⁺06], thin structure homogenised [CGM10a, CGM10b], or the jump smoothed over a few cells [TE03, TE06, KWR22]. Those methods choose the smoothing width from the grid and accept a second-order error component at the interface [TE06]; here the width is a property of the medium, smaller than the grid, and the aim is the scheme's full order through it. We found no prior instance of stencils on an unchanged equispaced or scattered node set whose basis is continued through a sub-grid smooth edge by ordinary differential equations, nor of the observation that the construction reduces to the jump stencils as $\delta \rightarrow 0$; the search is logged in the repository's `LITERATURE.md`. We ran the smoothing and averaging treatments on both schemes, as our implementation of each on one problem at one contrast (Sections 4 and 6): none reaches the seeds' order, the best of them is second order at a jump in one dimension, and the seeds lie 3 to 700 times below the best treatment there and 3 to 6 times below it on the scattered nodes. We claim nothing beyond that measurement.

Outline. Section 2 fixes the equations, the discretisations and the interface-aware stencils of the jump case that the seeds extend. Section 3 constructs the seeds in one dimension and proves what can be proved; Section 4 is the

one-dimensional experiment. Sections 5 and 6 do the same in two dimensions, flat edges first, then oblique incidence and curved edges. Section 7 records the limitations and what we would do next.

2. SETTING

2.1. One dimension. The one-dimensional problem is the two-way wave equation in first-order form on the periodic interval $[-1, 1)$, with u the particle velocity and f the stress,

$$(1) \quad \rho u_t = f_x, \quad f_t = K u_x, \quad K = \rho c^2,$$

where $c(x) > 0$ is the wave speed and $\rho(x) > 0$ the density. Eliminating one field gives a second-order operator for each,

$$(2) \quad u_{tt} = L_u u, \quad L_u = \frac{1}{\rho} \partial_x K \partial_x; \quad f_{tt} = L_f f, \quad L_f = K \partial_x \frac{1}{\rho} \partial_x.$$

The medium is a background material $(c, \rho) = (1, 1)$ with one layer $[x_1, x_2) = [0, \frac{1}{2})$ of $(c, \rho) = (2, 1)$, so the contrast in K is 4 and the impedance ρc doubles. With edge width $\delta > 0$ the two edges of the layer are tanh transitions,

$$(3) \quad s(x) = \frac{1}{2} \sum_{m \in \mathbb{Z}} \left[\tanh \frac{x - x_1 + 2m}{\delta} - \tanh \frac{x - x_2 + 2m}{\delta} \right],$$

$$c = c_{\text{bg}} + (c_{\text{layer}} - c_{\text{bg}}) s, \quad \rho \text{ likewise,}$$

summed over the periodic images so that the profile is smooth and periodic to rounding; $\delta = 0$ is the jump. For $\delta \ll h$ a stencil sees a jump between two of its nodes; for $\delta \gg h$ it sees a smooth function; the case in between is the subject of the paper.

The grid has n cell-centred nodes, $x_i = -1 + (i + \frac{1}{2})h$ with $h = 2/n$, and the semi-discrete system is

$$(4) \quad \rho_i \dot{u}_i = (D_f f)_i, \quad \dot{f}_i = K_i (D_u u)_i,$$

with $\rho_i = \rho(x_i)$, $K_i = K(x_i)$ and D_u, D_f two differentiation matrices with the sparsity of the centred five-point stencil. In the *naïve* scheme both are the standard fourth-order matrix, Fornberg's weights on the monomials [For88a]: the coefficients are sampled at the nodes and nothing else is done about the edge. In the *seed* scheme the rows whose stencils see a varying material are rebuilt as in Section 3, separately for u and for f because the two fields carry different continuity conditions; every other row is the standard one. Time stepping is the classical fourth-order Runge–Kutta method (RK4) with $\Delta t = 0.4 h / c_{\text{max}}$. Errors are relative ℓ^2 errors in the stress f at $t = 1$ against a reference solution (Section 4).

2.2. Two dimensions. The two-dimensional problem is the isotropic elastic wave equation in first-order form on the doubly periodic unit square, with velocities (u, v) and stresses $(\sigma_{xx}, \sigma_{xy}, \sigma_{yy})$,

$$(5) \quad \begin{aligned} \rho u_t &= \partial_x \sigma_{xx} + \partial_y \sigma_{xy}, & \partial_t \sigma_{xx} &= (\lambda + 2\mu) u_x + \lambda v_y, \\ \rho v_t &= \partial_x \sigma_{xy} + \partial_y \sigma_{yy}, & \partial_t \sigma_{xy} &= \mu (u_y + v_x), \\ & & \partial_t \sigma_{yy} &= \lambda u_x + (\lambda + 2\mu) v_y, \end{aligned}$$

with Lamé parameters $\lambda, \mu > 0$ and density $\rho > 0$, P- and S-wave speeds $c_p = \sqrt{(\lambda + 2\mu)/\rho}$ and $c_s = \sqrt{\mu/\rho}$, and $K = \lambda + 2\mu$ in the role of ρc^2 . (The implementation and the dissertation name the stresses f, g, h ; we write σ so that h can stay the node spacing.) The medium is a background $(\lambda, \mu, \rho) = (1, 1, 1)$ with a band $(4, 4, 2)$ between the two interfaces $y = y_0 + A \sin 2\pi x$, $y_0 = \frac{1}{4}$ and $\frac{1}{2}$, so c_p rises from $\sqrt{3}$ to $\sqrt{6}$ and c_s from 1 to $\sqrt{2}$ across each interface. The flat case is $A = 0$ and the curved case $A = 0.02$, the two geometries of the dissertation's elastic tests [Mar16]. With edge width $\delta > 0$ each interface becomes a tanh transition in the signed normal distance d to it, with the band's weight s summed over the periodic images in y ,

$$(6) \quad \begin{aligned} s &= \frac{1}{2} [1 + \tanh(d/\delta)], \\ (\lambda, \mu, \rho) &= (\lambda, \mu, \rho)_{\text{bg}} + [(\lambda, \mu, \rho)_{\text{band}} - (\lambda, \mu, \rho)_{\text{bg}}] s, \end{aligned}$$

so that λ, μ and ρ , and hence K , are linear in the weight while c_p and c_s are not. For a flat interface d is the vertical offset; for a curved one it is the true signed distance, through the foot point on the curve. Two edges a gap g apart reach only $\tanh(g/2\delta)$ of the contrast between them, so the band's width of $\frac{1}{4}$ bounds the edge widths we can pose; nothing here approaches that bound.

Node sets. A node set of n nodes is built as in [MF17a]: six fixed rows of nodes hug each interface, three on each side at normal distances $(\frac{1}{2} + j\frac{\sqrt{3}}{2})h$, $j = 0, 1, 2$, with alternate rows shifted by half a spacing so that the rows straddle the interface in a hexagonal pattern, and every other node is placed by a simulated electrostatic repulsion from a random start. The straddling rows were found empirically to be what keeps the interface-aware stencils stable, and we keep them for the seed stencils. Throughout, $h = n^{-1/2}$ denotes the mean spacing, and convergence rates are measured in it; the node counts are $n = 2500, 4900, 10,000$ and $19,600$, so h runs from 0.02 to 0.0071.

RBF-FD. For a stencil of s nodes $x_1, \dots, x_s$ about an evaluation node x_c , the weights w of a linear operator $\mathcal{L}$ solve the saddle-point system

$$(7) \quad \begin{bmatrix} A & P \\ P^\top & 0 \end{bmatrix} \begin{bmatrix} w \\ w_* \end{bmatrix} = \begin{bmatrix} \mathcal{L} \varphi(|x - x_i|)|_{x_c} \\ \mathcal{L} p_m(x)|_{x_c} \end{bmatrix},$$

with $A_{ij} = \varphi(|x_i - x_j|)$ for the Gaussian $\varphi(r) = e^{-(\varepsilon r)^2}$, $P_{im} = p_m(x_i)$ over the bivariate monomials p_m up to a chosen degree, evaluated in coordinates

scaled by the stencil radius, and w_* discarded. The constraint rows make the weights exact on the polynomials, which take over from the Gaussians under refinement, and the shape parameter is scaled to the local spacing, $\varepsilon = 0.4/d$ with d the distance from x_c to its third-nearest neighbour.¹ Away from the interfaces every stencil has 30 nodes and is exact on the polynomials of degree ≤ 4 ; stencils across a jump have 19 nodes and degree ≤ 3 , with the piecewise basis of Section 2.3. Two matrices D_x and D_y result, and the $5n \times 5n$ block operator of (5) is assembled from them with the material coefficients evaluated at each row's node. A small hyperviscosity $\gamma\Delta^3$ is added to every field to move the spurious eigenvalues of the RBF-FD operator into the left half-plane [FL11]: the Δ^3 stencils are built on the same nodes and the same basis as the derivative stencils, and $\gamma = \gamma_0 (h/h_0)^5$ with $\gamma_0 = 2.4 \times 10^{-11}$ at $h_0 = 0.02$, which keeps $\gamma\Delta^3$ proportional to $1/h$ like the wave operator so that one CFL number serves every n . Time stepping is RK4 with $\Delta t = 0.5 h/c_{p,\max}$, capped where necessary by the hyperviscosity spectrum.

Pulses and errors. The normal-incidence runs send a plane P-wave pulse, Gaussian in y with sharpness 15, from $y = 0.875$ in the background towards $-y$ through both interfaces; the true solution then has $u \equiv 0$, and the largest $|u|$ a scheme produces is a clean measure of what its stencils do at the edges. The oblique runs send a plane-wave train at 26.6° to the normal (direction $(1, 2)$), which mode-converts at each interface. Errors are relative ℓ^2 errors in v at $t = 1$ against the references described with each experiment in Section 6, together with the spurious u at normal incidence and the relative error in u otherwise.

2.3. Interface-aware stencils for a jump. The seeds extend the interface-aware stencils of [Mar16, MF17a], which we summarise in the form the later sections use; the derivations are in those references.

One dimension. Write (1) as $\partial_t(u, f)^\top = \mathcal{D}(u, f)^\top$ with the block operator

$$\mathcal{D} = \begin{bmatrix} 0 & \rho^{-1}\partial_x \\ K\partial_x & 0 \end{bmatrix}.$$

Across a material jump, u and f are continuous for all time, hence so are all their time derivatives, and $\partial_t^k(u, f)^\top = \mathcal{D}^k(u, f)^\top$ converts each of those conditions into a condition on spatial derivatives, with the material of the side it is evaluated on: $\mathcal{D}_L^k(u_L, f_L)^\top = \mathcal{D}_R^k(u_R, f_R)^\top$ at the jump for $k = 0, 1, 2, \dots$. Expanding u and f on each side as polynomials of degree four about the jump and imposing these conditions for $k \leq 4$ gives a square *continuity matrix* per side, $C_L a_L = C_R a_R$ on the coefficient vectors. Keeping the monomials on one side and translating them across with $C_R^{-1}C_L$ yields the *translated basis*: piecewise polynomials, one per monomial, that satisfy every continuity condition. A stencil whose nodes straddle both edges of a

¹The published description [MF17a] says the nearest neighbour; the original MATLAB and the present implementation use the third-nearest. The choice fixes a constant in εh and nothing else.

layer thinner than the stencil chains two translations. The stencil weights are then the ones that differentiate every basis function exactly at the evaluation point, a small dense solve per stencil, and the f basis is built by the same construction rather than from the u basis through the operator, which would lose an order. This is the scheme that Section 4 calls the jump case; the even continuity conditions relate a field to itself and the odd ones route through the other field, a pattern the seeds inherit.

Two dimensions. A stencil near an interface works in a local frame: origin at the nearest interface point, x' along the tangent and y' along the normal. The isotropic system (5) keeps its form under rotation, so in that frame the rotated fields obey the same equations and the interface is, locally, $y' = 0$. The velocities (u', v') and the tractions $(\sigma'_{x'y'}, \sigma'_{y'y'})$ are continuous across it for all time; the remaining stress $\sigma'_{x'x'}$ may jump. Expand (u', v') on each side in the monomials of (x', y') up to degree four, convert the time derivatives of the continuous quantities through powers of the block operator (even powers relate velocities to velocities, odd powers route through the stresses), and match the coefficients of the powers of x' at each order: a square continuity matrix per side, a translated velocity basis, truncated to degree three. The stress basis is generated from it by applying the stress rows of the operator on each side and dropping the three columns that come from rigid motions, 27 functions in all. The RBF-FD solve (7) then couples the fields: the stress rates need weights on u' and v' data jointly, the velocity rates weights on the three stresses jointly, with the Gaussian block repeated on the diagonal and the piecewise basis in place of P . The weights are rotated back to the laboratory frame, and the hyperviscosity stencils across the interface use the same piecewise basis, which the original work found necessary for accuracy. The interface is treated as locally flat in the frame; carrying curvature into the continuity conditions by expanding the interface in x' is possible [MF17a] but was not used by the original code and is not used here. This is the approximation we call zeroth order in the curvature, and the curved seeds of Section 5 make the same one.

2.4. Scope of the implementation. The experiments run on a Python reimplementations of the 2016–2017 methods (NumPy and SciPy, with a test for every numerical routine) that reproduces the constructions above and restructures the code. Of the original two-dimensional code it leaves out three things: the *triple stencils* that see both interfaces of a band thinner than one stencil (the band here is $\frac{1}{4}$ wide and none is needed), the variable Lamé parameters inside the band of the published test case (the band here is a constant material), and the finite-difference far field of the FD/RBF-FD hybrid (every stencil here is RBF-FD). Point sources and the Marmousi model are out of scope. In one dimension the implementation takes piecewise-constant materials, with the smooth edges of (3), and not the dissertation’s smoothly varying background, because the exact reference of the jump case, a ray sum, needs constant pieces; the published treatment of a wave speed

that varies smoothly up to an interface, a degree-four Taylor expansion of c about the interface point [Mar16, MF17a], is the Taylor route of Section 3.4 and is not ported. Everything is fourth order (FD4 in one dimension, degree four away from and degree three across the edges in two) at one material contrast; Section 7 returns to both restrictions.

3. SEEDS IN ONE DIMENSION

Fix a stencil of the grid of Section 2.1: nodes $x_1 < \dots < x_p$, $p = 5$ for FD4, and an evaluation point x_e , one of the nodes. The standard weights are the ones that differentiate the monomials $(x - x_e)^k$, $k < p$, exactly at x_e ; the interface-aware weights of Section 2.3 do the same for the translated basis. This section builds the basis for a smooth edge and shows that the two are its limits.

3.1. Seeds as the profiles of solutions polynomial in time. Consider the velocity field with its operator L_u of (2) and look for solutions of $u_{tt} = L_u u$ that are polynomial in time, $u(x, t) = \sum_{j=0}^J t^j g_j(x)$. Matching powers of t gives $(j+2)(j+1)g_{j+2} = L_u g_j$, so the two top coefficients lie in $\ker L_u$ and the $t = 0$ profile g_0 satisfies $L_u^m g_0 = 0$ for some m . With constant coefficients $L_u = c^2 \partial_x^2$ and $\ker \partial_x^{2m}$ is exactly the polynomials of degree $< 2m$: the profiles of time-polynomial solutions are what plays the role of polynomials for the operator. We fix a basis of them by anchoring at the evaluation point.

Definition 3.1 (Seeds). Let x_e be the evaluation point, $c_e = c(x_e)$ and $K_e = K(x_e)$. The *seeds* of the u field at x_e are the functions $\phi_0, \phi_1, \phi_2, \dots$ with

$$(8) \quad \begin{aligned} \phi_0 &= 1, & K\phi_1' &= K_e, & \phi_1(x_e) &= 0, \\ L_u \phi_k &= k(k-1)c_e^2 \phi_{k-2}, & \phi_k(x_e) &= \phi_k'(x_e) = 0 & (k \geq 2). \end{aligned}$$

The seeds of the f field are defined by the same chain with L_f in place of L_u and $\phi_1'/\rho = 1/\rho_e$ in place of the flux condition.

Each seed is the profile of one time-polynomial solution, and the normalisation is chosen so that $\phi_k = (x - x_e)^k$ when the coefficients are constant: $L_u(x - x_e)^k = c^2 k(k-1)(x - x_e)^{k-2}$, and the anchor conditions fix the two integration constants. Read one at a time, the chain says what each monomial becomes through the edge. The constant solution $u \equiv C$ gives ϕ_0 . A solution linear in time, $u = g_0 + tg_1$ with $u_t = g_1 = C$, forces $L_u g_0 = 0$, that is $Kg_0' = \text{const}$: the x -like seed ϕ_1 is the profile of *constant flux*, $\phi_1(x) = K_e \int_{x_e}^x d\xi/K(\xi)$, and through the other field this says that $f_t = Ku_x$ is constant in x . It is the flux that stays smooth through the edge, not the slope. A solution with $u_{tt} = C$ forces $L_u g_0 = C$, and $C = 2c_e^2$ with the anchor conditions gives the x^2 -like seed; here $C \neq 0$ matters, since $C = 0$ would only return $\ker L_u$. The cubic condition $u_{ttt} = C$ fixes the profile of u_t , not of u , so ϕ_3 is the profile whose second time derivative is

x -like, $L_u \phi_3 = 6c_e^2 \phi_1$; through the other field, $f_{ttt} = K(u_{tt})_x = 6c_e^2 K_e$ is constant. The even seeds are thus the conditions $\partial_t^k u = \text{const}$ and the odd seeds the conditions $\partial_t^k f = \text{const}$, the same alternation between the fields as in the continuity conditions of the jump case, where the even conditions relate a field to itself and the odd ones route through the other field.

First-order form. The chain is integrated as a first-order system that never differentiates the coefficients. For the u seeds the state is (ϕ_k, ψ_k) with $\psi_k = K \phi'_k$ the flux,

$$(9) \quad \phi'_k = \psi_k / K, \quad \psi'_k = \rho k(k-1) c_e^2 \phi_{k-2},$$

with initial data $\phi_0(x_e) = 1, \psi_1(x_e) = K_e$ and zero otherwise; for the f seeds the state is (ϕ_k, ψ_k) with $\psi_k = \phi'_k / \rho, \phi'_k = \rho \psi_k, \psi'_k = k(k-1) c_e^2 \phi_{k-2} / K$, and $\psi_1(x_e) = 1/\rho_e$. All p seeds of a field march together, from x_e outward to each side of the stencil, since the right-hand side of ϕ_k only involves ϕ_{k-2} . In the stencil coordinate $\xi = (x - x_e)/h_s$, with h_s the stencil half-width, the chain reads the same with ∂_x replaced by ∂_ξ , and then $\phi_k \approx \xi^k$: the interpolation matrix $A_{kj} = \phi_k(\xi_j)$ is a small perturbed Vandermonde matrix, and its condition number stays between 20 and 60 for δ from 10^{-5} to 1 in the setting of Section 2.1. Only ϕ_1 has a non-zero derivative at the anchor, so the weights solve $A\tilde{w} = e_1$ and $w = \tilde{w}/h_s$. Figure 1 shows the u seeds of one stencil for three edge widths.

3.2. Nested kernels and the jump as the limit. Only the span of the seeds enters the weights, and the spans are canonical. Since L_u sends each seed two steps down the chain, $\text{span}\{\phi_0, \dots, \phi_k\}$ is the $(k+1)$ -th space of

$$(10) \quad \{1\} \subset \ker L_u \subset \{L_u g = \text{const}\} \subset \ker L_u^2 \subset \{L_u^2 g = \text{const}\} \subset \dots,$$

where $\{L_u g = \text{const}\}$ is short for the space of g with $L_u g$ constant, so that a different anchor or normalisation changes the basis and not the space, and the weights not at all.

For a jump, read L_u in the weak sense: ϕ and the flux $K\phi'$ continuous across it. Then $\ker L_u$ is spanned by 1 and a piecewise-linear function whose slopes are in the ratio K_L/K_R , which is the translated x of Section 2.3, the function that keeps $\rho c^2 u_x$ continuous; $L_u \phi_2 = 2c_e^2$ gives $\phi_2'' = 2c_e^2/c^2$ on each side, which is the continuity of u_{tt} ; and so on up the chain, each condition $\partial_t^k u = \text{const}$ or $\partial_t^k f = \text{const}$ being one of the continuity conditions the translated basis was built from. The chain for a jump is therefore the translated Taylor basis, and the interface-aware stencils are the seed stencils at $\delta = 0$: one construction with two implementations, algebra for a jump and an ODE march for a smooth edge. The limit is approached at first order in δ . For a stencil of half-width 0.02 straddling an edge of the default contrast, the seed weights differ from the interface-aware weights by 22%, 10%, 0.98% and 0.098% of the largest weight at $\delta = 10^{-2}, 10^{-3}, 10^{-4}$ and 10^{-5} , for both fields, and equally for a stencil that straddles both edges of a

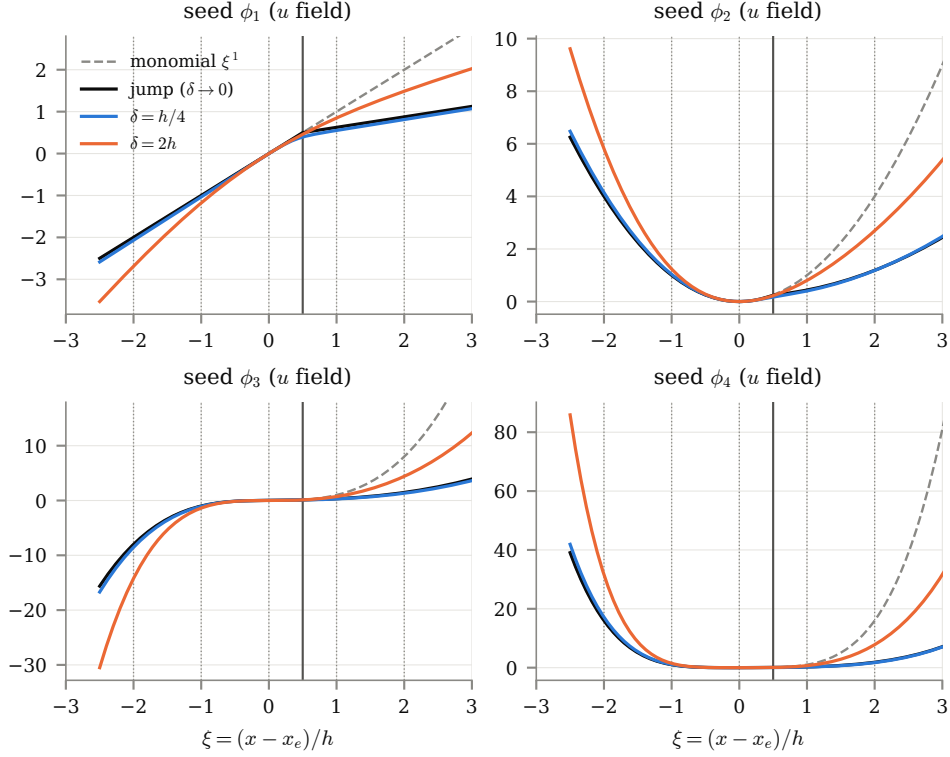


FIGURE 1. The u seeds $\phi_1, \dots, \phi_4$ of one five-point stencil whose evaluation point lies $h/2$ to the left of the lower edge of the layer of Section 2.1 (K from 1 to 4), for a jump, for $\delta = h/4$ and for $\delta = 2h$, against the monomials. The jump seeds are the translated Taylor basis of the interface-aware stencils: a slope ratio $K_L/K_R = 1/4$ for ϕ_1 , a curvature ratio $c_L^2/c_R^2 = 1/4$ for ϕ_2 . The seeds for $\delta = h/4$ are indistinguishable from them, the seeds for $\delta = 2h$ bend over the whole stencil, and all of them leave the monomials at the edge.

layer thinner than itself, the double-cross of the jump case, which the march handles by passing through both edges.

3.3. Well-posedness. The per-stencil solve $A\tilde{w} = e_1$ is nonsingular for every edge width, including the jump, and not only for the node sets we happened to use.

Proposition 3.2. *Let K and ρ be positive and continuous on an interval I containing x_e (or positive and piecewise continuous, with L_u read in the weak sense). Then for every k the seeds $\phi_0, \dots, \phi_k$ of Definition 3.1 form an extended complete Chebyshev system on I : interpolation in their span at any $k+1$ distinct points of I is uniquely solvable. The same holds for the f seeds.*

Proof. Each space in (10) is the kernel of an operator in Pólya form, a product of ∂_x and multiplications by positive functions:

$$(11) \quad \begin{aligned} \rho L_u^m &= \partial_x K \partial_x \frac{1}{\rho} \partial_x K \partial_x \cdots \partial_x K \partial_x & (2m \text{ derivatives}), \\ \partial_x L_u^m &= \partial_x \frac{1}{\rho} \partial_x K \partial_x \cdots \partial_x K \partial_x & (2m+1 \text{ derivatives}), \end{aligned}$$

with $\ker L_u^m = \ker(\rho L_u^m)$ the even spaces and $\{g : L_u^m g = \text{const}\} = \ker(\partial_x L_u^m)$ the odd ones; for L_f the roles of K and $1/\rho$ are swapped. The kernel of such an operator with positive weights $w_0, w_1, \dots$ has the nested basis $w_0, w_0 \int w_1, w_0 \int w_1 \int w_2, \dots$, and is an extended complete Chebyshev system on any interval where the weights are positive [P22, KS66, Cop71]; in particular it has the Haar property, so interpolation at distinct points in it is uniquely solvable. It remains to see that the seeds span these kernels. Their number matches the order of the operator, and they are independent because their jets at x_e are triangular: by induction on the chain, $(K\phi'_k)' = \rho k(k-1)c_e^2 \phi_{k-2}$ vanishes to order $k-2$ at x_e , so with $\phi_k(x_e) = \phi'_k(x_e) = 0$ the seed ϕ_k vanishes to order exactly k there. For the weak reading, the integrals in the nested basis are unaffected by finitely many jumps of the weights, and the argument is the same. $\square$

Interpolation in an extended complete Chebyshev system also has a Newton form with a remainder in which the operator that annihilates the space takes the place of the $(k+1)$ -th derivative (generalised divided differences, [M73]). For the five-point stencil the annihilator of $\text{span}\{\phi_0, \dots, \phi_4\}$ is $\partial_x L_u^2$, so the truncation error of the seed stencil on a function g is $O(h^4 \partial_x L_u^2 g)$. On a *solution* of the wave equation, $L_u u = u_{tt}$ and hence $\partial_x L_u^2 u = \partial_x \partial_t^4 u$, the x -derivative of another solution, which is bounded independently of δ : u_x changes by $O(1)$ across the edge whatever its width, and it is the flux Ku_x that is smooth. Standard FD4 has the error $O(h^4 u^{(5)})$ with $u^{(5)} \sim \delta^{-4}$ inside the edge. This is the prediction that Section 4 tests: standard FD4 converges at fourth order only once $h \lesssim \delta$ and looks low-order above that, with a knee at $h \approx \delta$; the seed stencils converge at fourth order with a constant that does not depend on δ .

Remark 3.3. The seeds are close relatives of two classical objects. Splines whose pieces lie in $\ker L$ for a differential operator L are the L-splines of [SV67], and the harmonic-mean coefficient $h/\int dx/K$ of the conservative difference schemes of [TS62] is the two-point version of ϕ_1 . Section 1.1 places the construction among them.

3.4. Why the Taylor route does not reach a sub-grid edge. The interface-aware construction of Section 2.3 can take smoothly varying coefficients: the continuity conditions are built from multiplication matrices formed from the Taylor coefficients of $1/\rho$ and K about the interface [Mar16, MF17a], and polynomial quasi-Trefftz spaces [IGMS23] do the same in general, with basis functions that satisfy the variable-coefficient equation to Taylor order at a point. For a sub-grid edge this route cannot work, and

the reason is not the algebra. The profile $\tanh(x/\delta)$ has poles at $\pm i\pi\delta/2$, so its Taylor series about the edge converges only within a distance $\pi\delta/2$ of it, while the stencil reaches $2h \gg \delta$; no polynomial degree that the stencil could carry represents the coefficients across it. The ODE march is what a sub-grid edge needs. For a resolved edge, $\delta \gg h$, the seed weights still differ from Fornberg’s at first order in h/δ , by 17% of the largest weight at $\delta = 8h$ for the default contrast, because the seeds are exact for a different five-dimensional space; both stencils are fourth order on solutions there, and Section 4 finds them indistinguishable end to end.

3.5. Stencil assembly and cost. The seed scheme rebuilds every row of D_u and D_f whose stencil nodes see different material values, and leaves every other row the standard one. For a $\tanh$ edge that region reaches about 19δ from each edge centre, where the tails round to the far-field value in double precision, plus the stencil’s reach. In each rebuilt row the $2p$ states of (9) for each field are integrated from $\xi = 0$ outward to each side with an explicit adaptive Runge–Kutta method of order eight, at relative tolerance 10^{-13} and absolute tolerance 10^{-15} , with segment boundaries at the edge centres and at $\pm 10\delta$ from them, so that the adaptive step never has to discover the edge in the middle of a long step across flat material. One march serves both fields and all p seeds, and a stencil that straddles both edges of a thin layer needs nothing further. The two $p \times p$ solves follow. In the reference implementation a march costs 5 to 15 ms, and the $n = 1600$ operator at $\delta = 0.01$ takes 2.3 s to build: seconds rather than milliseconds, fine for a study and not tuned. Two remarks on the range of δ . Since the seeds converge to the translated basis at first order in δ , an edge below about $10^{-5}h$ is better served by the interface-aware weights directly; and for δ of the order of the stencil or wider every row is rebuilt and the standard weights would have done as well (Section 3.4), so the rule of when to seed, settled in two dimensions in Section 6, applies here too.

4. ONE-DIMENSIONAL RESULTS

The experiment of this section is the one-dimensional test of the prediction that closes Section 3.3: standard FD4 loses its order through an edge it does not resolve and recovers it at $h \approx \delta$, while the seed stencils on the same grid are fourth order at every resolution with a constant that does not depend on δ .

Test problem. The medium is that of Section 2.1, the layer $[0, \frac{1}{2})$ with c from 1 to 2 at $\rho = 1$, so that K quadruples and the impedance doubles across each edge, with edges of width $\delta \in \{0, 0.0025, 0.01, 0.04\}$. The initial condition is a right-going Gaussian stress pulse, $f = \exp(-60(x+0.6)^2)$ and $u = -f/\rho c$, started in the background far enough from the layer that the tails of the reference below are clean. Its e^{-1} width is $2/\sqrt{60} \approx 0.26$, thirteen spacings on the coarsest grid, so the pulse is resolved on every grid and the edges are the only thing under test. The grids have $n = 100, 200, 400, 800$

and 1600 nodes, $h = 0.02$ down to 0.00125; for $\delta = 0.0025$ the ratio h/δ runs from 8 to $\frac{1}{2}$, for $\delta = 0.01$ from 2 to $\frac{1}{8}$, and for $\delta = 0.04$ every grid resolves the edge, $h/\delta \leq \frac{1}{2}$. Both schemes step with RK4 at $\Delta t = 0.4 h/c_{\max}$ to $t = 1$, by which time the pulse has crossed the layer: the transmitted pulse is in the background beyond the layer, the reflection from the lower edge is on its way back, and the reflection from the upper edge is still inside the layer (Figure 3). The error is the relative ℓ^2 error in f over the whole grid at $t = 1$.

References. For the jump the reference is the exact solution, a ray sum over the reflections and transmissions at the two interfaces. For $\delta > 0$ the coefficients are smooth and periodic and the reference is a Fourier pseudo-spectral discretisation of (1): derivatives in Fourier space, the coefficients applied pointwise, the same RK4 loop. It shares nothing with the finite-difference stencils under test, and the semi-discretisation conserves the energy $\int (f^2/K + \rho u^2) dx$ exactly, the Fourier derivative being skew-symmetric. Its resolution is set by the edge: on the interval of period 2, $\tanh(x/\delta)$ has poles at $\pm i\pi\delta/2$, so its Fourier coefficients decay like $e^{-\pi^2\delta m/2}$ in the mode number m , and the grid is the power of two that carries every mode down to 10^{-14} : 8192 nodes for $\delta = 0.0025$, 2048 for 0.01, and the floor of 1024 for 0.04. With $\Delta t = 5 \times 10^{-5}$ the reference is accurate to about 10^{-10} , below every error in Table 1 by more than two orders of magnitude, and is interpolated to the test grids trigonometrically.

The knee. Table 1 and Figure 2 give the errors and the rates between successive grids. For the jump the picture is the known one [Mar16, MF17a]: standard FD4 is first order and the interface-aware stencils are fourth order, 3.9×10^{-3} down to 6.2×10^{-8} . Through the edge of width $\delta = 0.0025$ standard FD4 converges at rates 1.3 and 2.2 while $h > \delta$, then drops by a factor 200 between $n = 400$ and $n = 800$, from $h = 2\delta$ to $h = \delta$: the knee. Over the two doublings past $h = 2\delta$ its error falls by a factor 850, an average rate of 4.9, unevenly (7.7, then 2.0), because the nodes sample the $\tanh$ at positions that move with n and so does the jump the stencil effectively sees; this is also why the rates before the knee are not the clean 1.0 of the jump. For $\delta = 0.01$ the knee sits between $n = 100$ and 200, again at $h \approx \delta$, and the rates after it are 4.0, 5.0 and 4.0. For $\delta = 0.04$, which every grid resolves, there is no knee and standard FD4 is fourth order throughout.

The seed stencils are fourth order at every n in every panel. Their errors through $\delta = 0.0025$ agree with the interface-aware errors of the jump case to three digits at every n , and through $\delta = 0.01$ to within 2%: the constant is the one of the jump construction, which is what Section 3.3 predicted, since the truncation error $O(h^4 \partial_x \partial_t^4 u)$ does not see the edge width. At $h = 8\delta$ the seeds are $18\times$ more accurate than standard FD4, at $h = 2\delta$ $400\times$, and at $h = \delta/2$, the end of the sweep for $\delta = 0.0025$, still $120\times$. Through the resolved edge $\delta = 0.04$ the two schemes agree to three digits at every n although their weights differ at first order in h/δ (Section 3.4): both are

TABLE 1. Relative ℓ^2 error in f at $t = 1$ against the reference, and the convergence rates between successive grids, for standard FD4 and the seed stencils (the interface-aware stencils at $\delta = 0$) on the same grids.

δ	scheme	$n = 100$	200	400	800	1600	rates
0 (jump)	standard FD4	7.1×10^{-2}	3.2×10^{-2}	1.58×10^{-2}	7.9×10^{-3}	3.9×10^{-3}	1.2, 1.0, 1.0, 1.0
0 (jump)	interface-aware	3.9×10^{-3}	2.5×10^{-4}	1.58×10^{-5}	9.9×10^{-7}	6.2×10^{-8}	4.0, 4.0, 4.0, 4.0
0.0025	standard FD4	7.2×10^{-2}	2.9×10^{-2}	6.3×10^{-3}	3.0×10^{-5}	7.4×10^{-6}	1.3, 2.2, 7.7, 2.0
0.0025	seeds	3.9×10^{-3}	2.5×10^{-4}	1.57×10^{-5}	9.9×10^{-7}	6.2×10^{-8}	4.0, 4.0, 4.0, 4.0
0.01	standard FD4	3.0×10^{-2}	5.6×10^{-4}	3.5×10^{-5}	1.05×10^{-6}	6.5×10^{-8}	5.7, 4.0, 5.0, 4.0
0.01	seeds	3.9×10^{-3}	2.5×10^{-4}	1.55×10^{-5}	9.7×10^{-7}	6.1×10^{-8}	4.0, 4.0, 4.0, 4.0
0.04	standard FD4	3.5×10^{-3}	2.2×10^{-4}	1.41×10^{-5}	8.8×10^{-7}	5.5×10^{-8}	4.0, 4.0, 4.0, 4.0
0.04	seeds	3.5×10^{-3}	2.2×10^{-4}	1.40×10^{-5}	8.8×10^{-7}	5.5×10^{-8}	4.0, 4.0, 4.0, 4.0

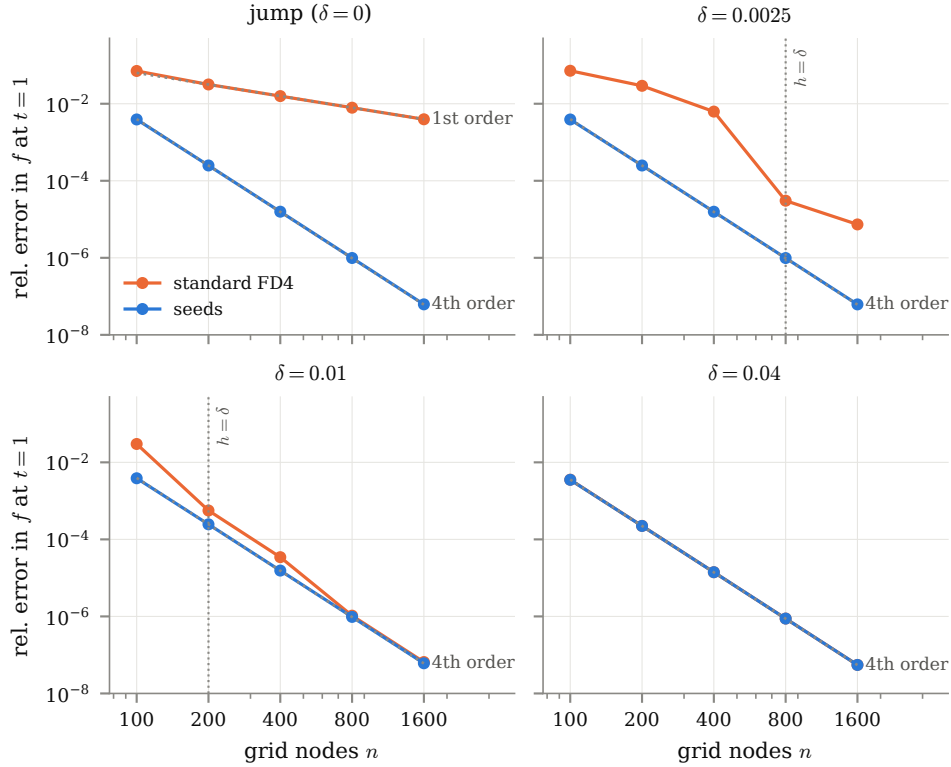


FIGURE 2. The errors of Table 1 against n for the four edge widths, with first- and fourth-order guides and, for $\delta > 0$, the grid at which $h = \delta$. Standard FD4 has its knee there; the seed stencils lie on one fourth-order line in every panel.

exact on a five-dimensional space that approximates the solution to fourth order, and on a resolved edge it does not matter which.

Truncation error. The same story is told without the time stepping by the local truncation error of the two differentiation matrices on the reference solution's profile at $t = 1$, here for the narrower pulse of the dissertation (sharpness 600), as the largest error in u_x over the rebuilt rows relative to $\max |u_x|$. Through $\delta = 0.0025$ the seed rows have errors 9.9×10^{-3} , 1.2×10^{-3} , 9.1×10^{-5} , 5.8×10^{-6} and 3.6×10^{-7} at $h = 8\delta$, 4δ , 2δ , δ and $\delta/2$, successive ratios 8.5, 13, 16 and 16, fourth order from $h = 8\delta$ on; standard FD4 on the same rows falls only from 1.5×10^{-2} to 2.2×10^{-4} over the same five grids, and not monotonically. Through $\delta = 0.01$ the seed rows give 2.2×10^{-2} down to 2.3×10^{-7} on the same five grids: the same constant.

The coarse-grid picture. Figure 3 is the run at $n = 100$ through edges of width $\delta = 0.0025 = h/8$, so that both edges of the layer fall between nodes. Standard FD4 trails a grid-scale sawtooth of about 4% of the pulse height over the whole interval, ahead of the pulses as well as behind them: the highest modes of the FD4 stencil have negative group velocity, so the noise born at the edges outruns the physical waves. The seed stencils sit on the reference with a maximum error of 0.3% of the pulse height, which is the dispersion error of interior FD4 on the pulse itself.

Conditioning and stability. The 5×5 interpolation matrices of the rebuilt rows are the perturbed Vandermonde matrices of Section 3.1, with condition numbers between 20 and 60 for δ from 10^{-5} to 1; nothing in the sweep approaches the range of δ where the interface-aware weights should be used directly. The semi-discrete operator of the seed scheme has no growing modes to speak of: the largest real part of its spectrum is 8×10^{-8} at $\delta = 0.005$ on 400 nodes and about 10^{-13} in the other cases examined, and RK4 at $\Delta t = 0.4 h/c_{\max}$ was stable in every run of the sweep. The runs cost seconds (Section 3.5). Nothing was tried at sharper contrasts, for layers thinner than the 19δ tails of an edge, or at orders above four; the construction does not change for any of them, and Section 7 lists them among the limitations.

The standing alternative: the standard scheme on a changed medium. In seismic and electromagnetic finite differences an interface is routinely handled by changing the medium rather than the stencil (Section 1.1). Tornberg and Engquist [TE06] derive the change for the system $p_t = au_x$, $u_t = bp_x$ with $a = -\rho c^2$ and $b = -1/\rho$: $1/a$ and $1/b$ linear across the one cell centred on the jump, that is, the compliance $1/K$ and the density replaced by their cell means, the harmonic and arithmetic means of the Moczo line [MKV⁺02]; their Theorem 1 gives second order for the Yee scheme, regularising a itself instead of $1/a$ stays first order with a smaller constant, and their fourth-order staggered scheme reaches second order only with its temporal correction terms masked near the jump, terms the collocated scheme here

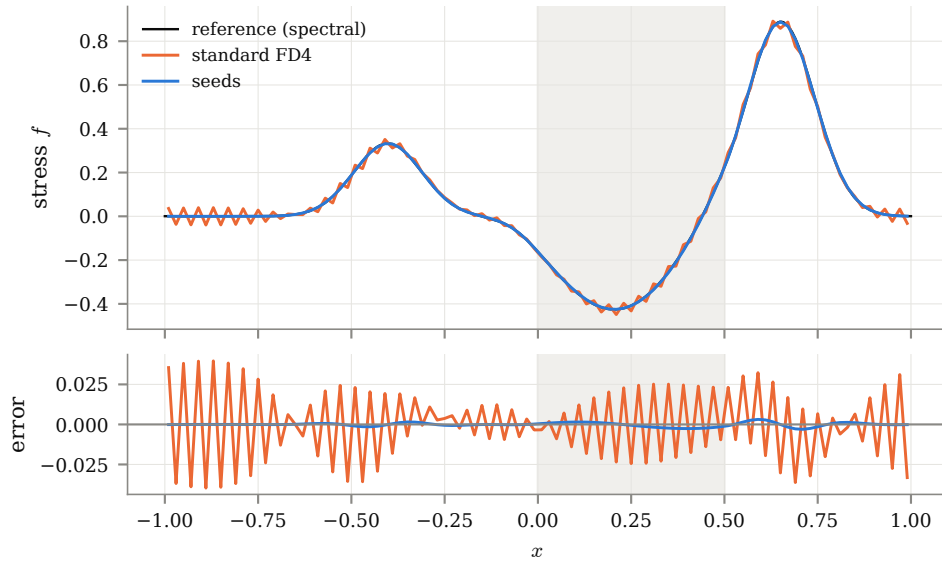


FIGURE 3. Stress f at $t = 1$ on 100 nodes through edges of width $\delta = h/8$ (the layer shaded), against the spectral reference, and the pointwise errors of the two schemes. The transmitted pulse has left the layer, the reflection from the lower edge is travelling back and the reflection from the upper edge is still inside; standard FD4 carries a grid-scale sawtooth everywhere, the seed stencils only the interior dispersion error.

does not have. Koene, Wittsten and Robertsson [KWR22] instead band-limit the density and the compliance to the grid Nyquist with a windowed low-pass filter. We ran both on the scheme of this section: the same FD4 and RK4, grids, pulse and true- δ reference as Table 1, with the coefficients the standard scheme samples replaced by their cell means over one cell and over two (the layer's edges sit on cell boundaries of every grid here, so the one-cell ramp of a jump ends exactly at the nearest nodes and changes nothing at $\delta = 0$; a staggered grid always has a coefficient position within $h/4$ of a jump, a collocated one does not), by the band-limited coefficients (cutoff $1.1\pi/h$, Hanning window of half-width $2.5h$, their filter applied to the true profile), and, as the coarse bound, by the true profile with its edge widened to $\max(\delta, h)$ and $\max(\delta, 2h)$, which blends c and ρ and is the case they warn about. Each treatment is applied at its prescribed width at every n : the sources have no δ and state no switch. Table 2 and Figure 4 give the result.

At the jump no coefficient treatment lifts the collocated scheme's first order except the two-cell mean, which is second order at every n , $5.5\times$ below the sampled medium at $n = 100$ and $87\times$ at $n = 1600$; the interface-aware stencils are $3.3\times$ below it on the coarsest grid and $730\times$ on the finest. The

band-limited medium keeps first order at $0.8\times$ the constant, the widened edges at $1.4\times$ and $2.7\times$. Through the unresolved edge $\delta = 0.0025$ the one-cell mean and the band-limited medium cut the sampled error by 1.25 to $1.6\times$ while $h > \delta$ and follow it through the knee; the two-cell mean stays second order, 5.7 to $11\times$ below the sampled medium while $h \geq 2\delta$ and 4.7 to $4.9\times$ above it from $h = \delta$ on, where the sampled scheme is fourth order and the averaged medium is not the true one. The seeds are $3.2\times$ below the two-cell mean at $h = 8\delta$, $37\times$ at $h = 2\delta$, $146\times$ at $h = \delta$ and $580\times$ at $h = \delta/2$: the gap widens toward the knee. Widening the edge is worse than sampling it wherever it acts (1.03 to $1.7\times$ for one cell, 2.2 to $5.3\times$ for two) and equal to it once the true edge is the wider one. On a resolved edge every treatment at its prescribed width is second order and falls behind the fourth-order sampled scheme, by 47 to $530\times$ at $n = 1600$ through $\delta = 0.01$ and 0.04 : the second-order component the literature describes, measured on this scheme. The treatments therefore need the same switch as the seeds, applied while $h > \delta$ and not otherwise, which their sources, written for jumps, do not state; with it the best of them is second order through the edge at no cost per stencil, and the seeds are fourth order at one ODE march per rebuilt row. What this licenses is stated in Section 1.1: one scheme, one contrast, one pulse, our implementation of each treatment on a collocated grid, against sources that analyse and run them on staggered grids.

5. SEEDS IN TWO DIMENSIONS

The chain of Section 3.1 is dimension-free: for any operator $\mathcal{L}$ the nested spaces $\{1\} \subset \ker \mathcal{L} \subset \{\mathcal{L}g = \text{const}\} \subset \ker \mathcal{L}^2 \subset \dots$ are what they were. What is not free is the choice of basis. In two dimensions $\ker \mathcal{L}$ is infinite-dimensional, and a space of “polynomial-like functions of degree $\leq q$ ” that reduces to the polynomials for constant coefficients needs a rule for which $\mathcal{L}$ -harmonic functions play the role of $1, x, y, xy, x^2 - y^2, \dots$: each harmonic polynomial needs a continuation through the edge that looks like it away from the edge. For a straight edge that rule is an ansatz, and it turns the problem back into ordinary differential equations in the normal coordinate. This section constructs the seeds of the elastic system for a straight edge, assembles them into stencils, and says what a curved edge does to the construction.

5.1. Design: a straight edge. A stencil that sees an edge works in the local frame of Section 2.3: origin at the nearest point of the edge centre, x along the tangent and y along the normal, the rotated fields obeying (5) unchanged. In this section (x, y) are those local coordinates and (u, v) and σ the rotated fields, the primes of Section 2.3 dropped. The material depends on y alone: exactly, for a flat edge, and for a curved one in the approximation

TABLE 2. The standard scheme on a treated medium (the standing alternative) against the standard scheme on the sampled medium and the seeds: relative ℓ^2 error in f at $t = 1$ against the true medium's reference, and the rates between successive grids. The cell means and the band-limited medium are applied at their prescribed width at every n ; the widened edges of the text are omitted, they never beat the sampled medium.

δ	medium, scheme	$n = 100$	200	400	800	1600	rates
0 (jump)	standard FD4	7.1×10^{-2}	3.2×10^{-2}	1.58×10^{-2}	7.9×10^{-3}	3.9×10^{-3}	1.2, 1.0, 1.0, 1.0
0 (jump)	cell mean, h	7.1×10^{-2}	3.2×10^{-2}	1.58×10^{-2}	7.9×10^{-3}	3.9×10^{-3}	1.2, 1.0, 1.0, 1.0
0 (jump)	cell mean, $2h$	1.30×10^{-2}	2.9×10^{-3}	7.3×10^{-4}	1.81×10^{-4}	4.5×10^{-5}	2.1, 2.0, 2.0, 2.0
0 (jump)	band-limited	5.7×10^{-2}	2.5×10^{-2}	1.26×10^{-2}	6.3×10^{-3}	3.1×10^{-3}	1.2, 1.0, 1.0, 1.0
0 (jump)	interface-aware	3.9×10^{-3}	2.5×10^{-4}	1.58×10^{-5}	9.9×10^{-7}	6.2×10^{-8}	4.0, 4.0, 4.0, 4.0
0.0025	standard FD4	7.2×10^{-2}	2.9×10^{-2}	6.3×10^{-3}	3.0×10^{-5}	7.4×10^{-6}	1.3, 2.2, 7.7, 2.0
0.0025	cell mean, h	5.7×10^{-2}	1.90×10^{-2}	3.9×10^{-3}	4.3×10^{-5}	1.02×10^{-5}	1.6, 2.3, 6.5, 2.1
0.0025	cell mean, $2h$	1.26×10^{-2}	2.6×10^{-3}	5.9×10^{-4}	1.44×10^{-4}	3.6×10^{-5}	2.3, 2.1, 2.0, 2.0
0.0025	band-limited	5.3×10^{-2}	1.84×10^{-2}	3.8×10^{-3}	3.0×10^{-5}	5.9×10^{-6}	1.5, 2.3, 7.0, 2.4
0.0025	seeds	3.9×10^{-3}	2.5×10^{-4}	1.57×10^{-5}	9.9×10^{-7}	6.2×10^{-8}	4.0, 4.0, 4.0, 4.0
0.01	standard FD4	3.0×10^{-2}	5.6×10^{-4}	3.5×10^{-5}	1.05×10^{-6}	6.5×10^{-8}	5.7, 4.0, 5.0, 4.0
0.01	cell mean, h	2.0×10^{-2}	7.6×10^{-4}	1.45×10^{-4}	3.5×10^{-5}	8.7×10^{-6}	4.7, 2.4, 2.0, 2.0
0.01	cell mean, $2h$	1.07×10^{-2}	2.3×10^{-3}	5.6×10^{-4}	1.39×10^{-4}	3.5×10^{-5}	2.2, 2.0, 2.0, 2.0
0.01	band-limited	2.0×10^{-2}	5.3×10^{-4}	5.9×10^{-5}	1.42×10^{-5}	3.6×10^{-6}	5.2, 3.2, 2.1, 2.0
0.01	seeds	3.9×10^{-3}	2.5×10^{-4}	1.55×10^{-5}	9.7×10^{-7}	6.1×10^{-8}	4.0, 4.0, 4.0, 4.0
0.04	standard FD4	3.5×10^{-3}	2.2×10^{-4}	1.41×10^{-5}	8.8×10^{-7}	5.5×10^{-8}	4.0, 4.0, 4.0, 4.0
0.04	cell mean, h	4.2×10^{-3}	5.0×10^{-4}	1.04×10^{-4}	2.5×10^{-5}	6.3×10^{-6}	3.1, 2.3, 2.0, 2.0
0.04	cell mean, $2h$	7.9×10^{-3}	1.66×10^{-3}	4.0×10^{-4}	1.01×10^{-4}	2.5×10^{-5}	2.2, 2.0, 2.0, 2.0
0.04	band-limited	3.5×10^{-3}	2.5×10^{-4}	4.1×10^{-5}	1.02×10^{-5}	2.6×10^{-6}	3.8, 2.6, 2.0, 2.0
0.04	seeds	3.5×10^{-3}	2.2×10^{-4}	1.40×10^{-5}	8.8×10^{-7}	5.5×10^{-8}	4.0, 4.0, 4.0, 4.0

of Section 5.4. Eliminating the stresses from (5), with $K = \lambda + 2\mu$,

$$(12) \quad \begin{aligned} \rho u_{tt} &= \partial_x [K u_x + \lambda v_y] + \partial_y [\mu(u_y + v_x)], \\ \rho v_{tt} &= \partial_x [\mu(u_y + v_x)] + \partial_y [\lambda u_x + K v_y], \end{aligned}$$

and $\mathcal{L}(u, v)$ denotes the right-hand side divided by ρ . The three brackets are the stress rates of (5), and the two that carry ∂_y are the tractions on the edge, σ_{xy} and σ_{yy} , which are what stays continuous across a jump. That is the 2-D form of the observation that drove the 1-D construction: the flux $K\phi'$, not the slope, is the smooth quantity through the edge.

Proposition 5.1 (Reduction to the normal coordinate). *Let λ, μ, ρ depend on y only, and let $u = \sum_{j=0}^q a_j(y) x^j$ and $v = \sum_{j=0}^q b_j(y) x^j$. Write the x^j -coefficients of the three stress rates as*

$$(13) \quad \begin{aligned} F_j &= K(j+1) a_{j+1} + \lambda b'_j, & \tau_j &= \mu(a'_j + (j+1) b_{j+1}), \\ \nu_j &= \lambda(j+1) a_{j+1} + K b'_j, \end{aligned}$$

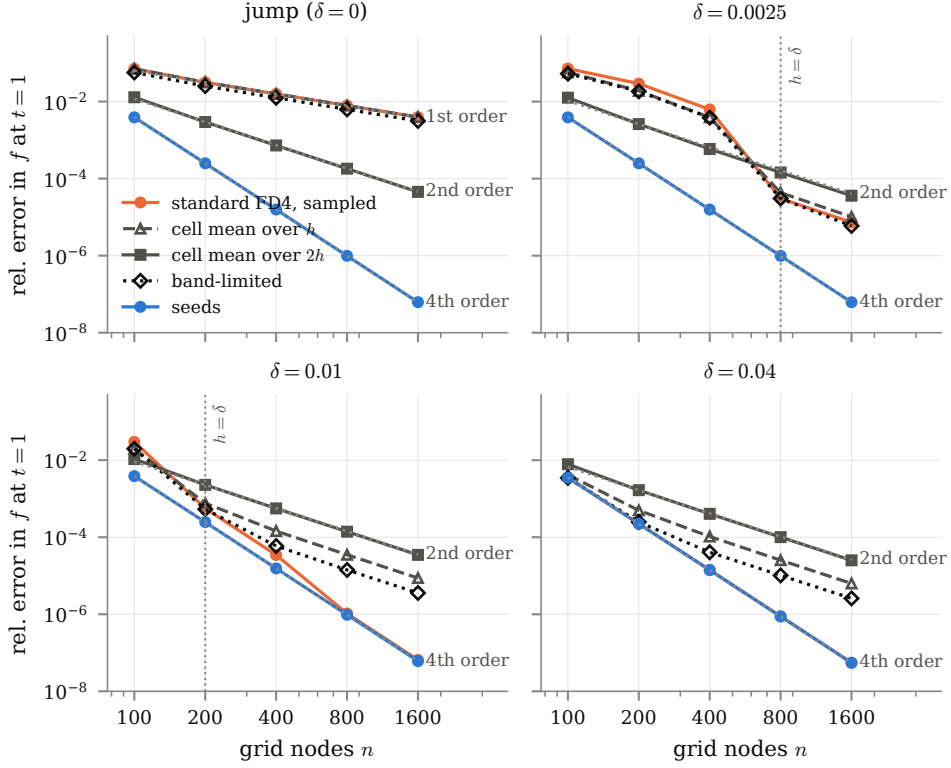


FIGURE 4. Table 2 drawn: error against n for the sampled medium, the three treated media and the seeds, per edge width. The two-cell mean is second order everywhere and never sees the knee; the one-cell mean and the band-limited medium track the sampled scheme; the seeds are fourth order at every n .

for σ_{xx} , σ_{xy} and σ_{yy} respectively, with $a_j = b_j = 0$ for $j > q$. Then $\rho\mathcal{L}(u, v)$ is a polynomial in x of degree $\leq q$ whose x^j -coefficients are

$$(14) \quad \rho\mathcal{L}_{1,j} = (j+1)F_{j+1} + \tau'_j, \quad \rho\mathcal{L}_{2,j} = (j+1)\tau_{j+1} + \nu'_j.$$

In particular the coefficients at level j involve (a, b) at levels j , $j+1$ and $j+2$ only, so an equation $\mathcal{L}(u, v) = (\text{given})$ is a triangular system, from the top x -degree down, of two-component second-order ODEs in y for the pairs (a_j, b_j) .

Proof. Each bracket in (12) is a polynomial in x with coefficient functions of y , as displayed in (13); ∂_x shifts the index and multiplies by it, ∂_y differentiates the coefficient, and (14) follows by reading off. Expanding F_{j+1} gives $\rho\mathcal{L}_{1,j} = (j+1)(j+2)Ka_{j+2} + (j+1)\lambda b'_{j+1} + \tau'_j$, which reaches level $j+2$; τ'_j and ν'_j involve a''_j , b''_j and b'_{j+1} , a'_{j+1} , and nothing lower. $\square$

First-order form. As in one dimension, the chain is integrated as a first-order system that never differentiates the coefficients, and the tractions are the flux variables. Solving (13) for the slopes and (14) for the derivatives of the tractions, the state at level j is $(a_j, b_j, \tau_j, \nu_j)$ with

$$(15) \quad \begin{aligned} a'_j &= \tau_j / \mu - (j+1) b_{j+1}, \\ b'_j &= (\nu_j - \lambda(j+1) a_{j+1}) / K, \\ \tau'_j &= \rho R_{1,j} - K(j+2)(j+1) a_{j+2} - \lambda(j+1) b'_{j+1}, \\ \nu'_j &= \rho R_{2,j} - (j+1) \tau_{j+1}, \end{aligned}$$

where b'_{j+1} in the third line is the second line at level $j+1$ and (R_1, R_2) is the right-hand side the chain prescribes, next. The coefficients λ, μ, ρ are only ever evaluated. The pair (τ_j, ν_j) is the 2-D twin of $\psi_k = K\phi'_k$ in (9).

5.2. The elastic seeds. With constant coefficients $\mathcal{L}$ maps a pair of monomials to pairs of lower degree,

$$(16) \quad \begin{aligned} \mathcal{L}(x^a y^b, 0) &= \frac{1}{\rho} \begin{pmatrix} Ka(a-1)x^{a-2}y^b + \mu b(b-1)x^a y^{b-2} \\ (\lambda + \mu)abx^{a-1}y^{b-1} \end{pmatrix}, \\ \mathcal{L}(0, x^a y^b) &= \frac{1}{\rho} \begin{pmatrix} (\lambda + \mu)abx^{a-1}y^{b-1} \\ \mu a(a-1)x^{a-2}y^b + Kb(b-1)x^a y^{b-2} \end{pmatrix}, \end{aligned}$$

that is, a matrix C over the $2m$ monomial pairs e (m monomials to degree q , the u -block then the v -block), $\mathcal{L}e = \sum_{e'} C[e', e] e'$, which lowers the degree by two and is therefore strictly triangular in degree. It is the (u, v) block of the square of the block operator of (5) on polynomial coefficients, the same object the continuity matrices of Section 2.3 are built from.

Definition 5.2 (Elastic seeds of a straight edge). Let the evaluation node be at (x_e, y_e) in the local frame and let C be the matrix (16) for the material at y_e . The *seeds* S_e , one per monomial pair e , are the solutions of

$$(17) \quad \mathcal{L}S_e = \sum_{e'} C[e', e] S_{e'}$$

of the form of Proposition 5.1 in $(x - x_e, y - y_e)$, with the monomial's jet as initial data at $y = y_e$: for $e = (x^\alpha y^\beta, 0)$, $a_\alpha(y_e) = 1$ if $\beta = 0$, $a'_\alpha(y_e) = 1$ if $\beta = 1$, and every other a_j, b_j and slope zero at y_e , with the mirror image for $e = (0, x^\alpha y^\beta)$; the tractions at y_e follow from the jet and the material there through (13).

This is Definition 3.1 read in two dimensions: $L_u \phi_k = k(k-1)c_e^2 \phi_{k-2}$ is (17) with C the 1-D chain matrix, and the anchor conditions are the jet. The seeds of the lower monomials stand on the right, not the monomials, which is what makes the system consistent: by the triangularity of C a seed only ever needs seeds two degrees down, and all $2m$ of them march together as one linear system. With constant coefficients the right-hand side of (17) is $\mathcal{L}$ of the monomial itself, by induction on the degree, and the linear ODE

with that data has one solution, the monomial: the seeds are the monomials, to 2.5×10^{-14} in the implementation's test. For the seed of $(x^\alpha y^\beta, \cdot)$ only the levels $j \leq \alpha$ are ever non-zero, since nothing above them is driven, so a seed keeps the x -degree of its monomial and the medium enters through the y -structure alone. Two seeds are in closed form through any edge: $(y, 0)$ becomes $(\int_{y_e}^y \mu_e/\mu \, d\eta, 0)$ and $(0, x)$ becomes $(\int_{y_e}^y (\mu_e/\mu - 1) \, d\eta, x)$, both with the constant shear traction μ_e and no other stress. That is the twin of $\phi_1 = K_e \int d\xi/K$: the traction is what stays smooth, not the slope.

The system for degree three. The stencils across an edge are exact on velocities to degree $p = 3$, and the stress basis is generated from velocities one degree higher, as in the jump case, so the seeds are marched for the monomials to degree $q = 4$: 15 monomials per component, 30 seeds, x -degrees 0 to 4, and five levels of $(a_j, b_j, \tau_j, \nu_j)$ per seed, a linear system of 600 states. It is integrated with the same explicit eighth-order Runge–Kutta method and tolerances as in one dimension, from the anchor outward to each side, restarted at every node's normal coordinate so that the node values are integrated rather than interpolated, and at the edge centres and their $\pm 10\delta$ flanks. The flanks matter here as they did in one dimension: a single solve without them differs from the segmented march by 4×10^{-10} at $\delta = 10^{-4}$, where the adaptive step has to discover the edge.

Stress seeds and rigid motions. The stress seed of a velocity seed has the components $\sigma_{xx} = \sum_j F_j x^j$, $\sigma_{xy} = \sum_j \tau_j x^j$ and $\sigma_{yy} = \sum_j \nu_j x^j$, read off the marched state and the flux variables, with b'_j in F_j taken from the second line of (15), so nothing is differentiated numerically. As in the jump construction (Section 2.3), the constants in u and v have no stress and the shear pair $(y, 0)$, $(0, x)$ share theirs, through any edge and not only for constant coefficients, so the same three columns are dropped: 27 stress seeds from the 30 velocity seeds, and the 20 velocity seeds of degree ≤ 3 for the velocity block, the dissertation's counts.

Anchor, scaling and jets. The seeds are anchored at the evaluation node in both coordinates and scaled by $r_{\max}$, the distance to the farthest node of the stencil, so that $S_e \approx X^a Y^b$ in $(X, Y) = (x - x_e, y - y_e)/r_{\max}$ and the material is sampled at $y_e + r_{\max} Y$. The jump basis of Section 2.3 keeps its origin at the interface point. Moving the origin in x is a change of basis within the same span and changes no weight, and the two conventions are reconciled that way, so the jump path is untouched. Where the *normal* coordinate is anchored is not a change of basis, because that is where the jet is imposed and the material is frozen in C ; there the seeds follow the lesson of one dimension, that anchoring at the evaluation point is what keeps the interpolation block conditioned. At the anchor the velocity seeds have the monomials' first derivatives (only X and Y have one), and the derivatives of the stress seeds come from the state and the right-hand side of (15) at $Y = 0$, so the elastic rates of every basis function at the evaluation node are known exactly. One of the implementation's tests is the identity this implies:

the velocity rate of a stress seed at the anchor, $(\partial_X \sigma_{xx} + \partial_Y \sigma_{xy})/\rho_e$, is the constant term of the seed's column of C .

Checks. Three properties were checked on a 19-node stencil of a 400-node set ($h = 0.05$) on the row nearest the lower interface, the anchor $h/2$ into the band or $h/2$ into the background. *Jump limit:* as $\delta \rightarrow 0$ the state passes through the edge unchanged, which is velocity and traction continuity, and on the far side the chain holds with the anchor's frozen coefficients, which is what the continuity matrices encode; so the seeds should tend to the translated basis of the jump case re-expanded about the anchor, and they do, at first order in δ : the largest difference relative to the largest basis value is 1.0 to 2.3% at $\delta = 10^{-3}$ and falls tenfold per decade to 1.0 to 2.4×10^{-4} at $\delta = 10^{-5}$, for the velocity and the stress basis and with the anchor on either side. *Residual:* every seed, evaluated as a function of (x, y) on a 25×401 tensor grid through a $\delta = 0.01$ edge and pushed through the elastic operator by eighth-order finite differences in both variables, satisfies (17) with relative residual 2×10^{-12} ; the test uses none of the module's coefficient bookkeeping, and the residual rises to 0.23 if the factor $(j+2)(j+1)$ in (15) is wrong and to 0.06 if the $\lambda(j+1)b'_{j+1}$ term is dropped. *Conditioning:* the 2-norm condition numbers of the velocity block (38×20 : two components at 19 nodes against 20 seeds) and the stress block (57×27) are 17 to 34 and 24 to 30 for δ from h down to $h/5000$, against 34 and 30 for the polynomial jump basis re-expanded about the same anchor, and they tend to those values as $\delta \rightarrow 0$. No Chebyshev-system argument is available in two dimensions, where interpolation on scattered nodes can be singular even for polynomials. The saddle-point solve is nonsingular when the Gaussian block is positive definite, which it is, and the augmentation has full column rank; for the seeds the measured conditioning is the evidence for the latter, and it is all we have.

5.3. Stencil assembly. *Which rows.* The seed scheme rebuilds the rows of every node whose 19 nearest nodes see different values of λ , μ or ρ , the 1-D rule of Section 3.5 on the stencil's footprint: with exact inequality that reaches about 19δ from each edge centre plus the stencil's radius, and a relative tolerance on the spread trims the tails (the trim of Section 6). The nearest interface, by vertical offset, sets the frame of a rebuilt row; the material profile along its normal carries both edges of the band, so a stencil that sees both is marched through both, and the triple stencils of the jump case are not needed.

Elastic rows. The rebuilt elastic rows are the coupled saddle-point solves of Section 2.3 with the seeds in place of the piecewise polynomials: the Gaussian block on the 19 nodes is unchanged, the augmentation is the 38×20 block of velocity-seed values and the 57×27 block of stress-seed values, the right-hand sides on the seed columns are the elastic rates of (5) at the evaluation node, from the jets, and the material in those rates is the anchor material, which is the node's own. The weights are rotated back to the

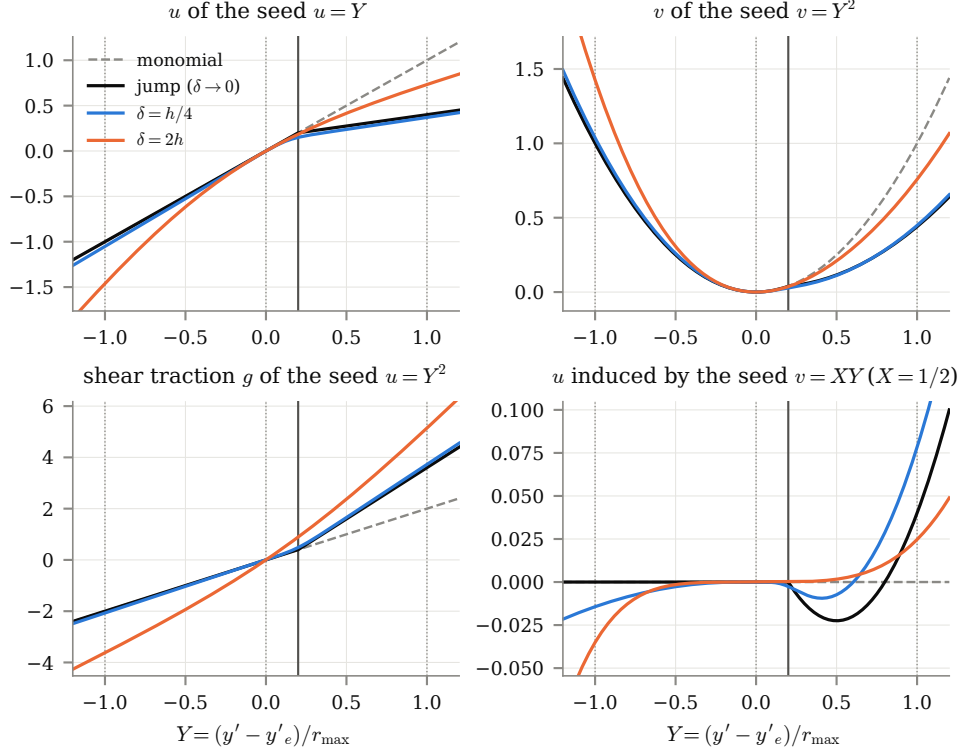


FIGURE 5. Cross-sections along the normal through the anchor of four elastic seeds of a 19-node stencil whose evaluation node lies $h/2$ below the lower edge of the band of Section 2.2 (λ, μ from 1 to 4, ρ from 1 to 2), in the stencil coordinate $Y = (y - y_e)/r_{\max}$ with $r_{\max} = 2.5h$, for the jump limit (computed at $\delta = 10^{-6}$), $\delta = h/4$ and $\delta = 2h$, against the monomials of the anchor material (dashed): the u of the seed $u = Y$; the v of the seed $v = Y^2$; the shear traction of the seed $u = Y^2$ (labelled g , the implementation's name for the rate of σ_{xy}); and the u induced by the seed $v = XY$ at $X = \frac{1}{2}$, which is zero for the monomial and is the tangential coupling that the ablation of Section 6 removes. The vertical line is the edge centre and the dotted lines the stencil's reach $|Y| \leq 1$. As in Figure 1, the $\delta = h/4$ seeds are close to the jump limit and the $\delta = 2h$ seeds bend over the whole stencil.

laboratory frame as before. Every stencil's seeds are its own, even on a flat edge, because the jet and the frozen C live at its anchor.

Hyperviscosity rows. The Δ^3 rows of the same nodes are built on the naive footprint of 30 nodes, with the same seeds annihilated: the saddle-point solve imposes $\Delta^3 S_e = 0$ on every seed column, as it imposes $\Delta^3 p = 0$ on

the polynomial augmentation of a standard stencil. That is a second march per stencil, over the larger radius. Two things decided the footprint, both measured in Section 6. A 19-node stencil carrying 20 coupled constraints has 18 degrees of freedom left for the Gaussian part of a sixth-order operator, and what comes out is barely a Δ^3 : on the 900-node set the extreme eigenvalue of its Δ^3 block is -82 against -151 for the standard rows, half the damping, and the operator built that way is unstable at the standard γ once the seeded region, 19δ plus the stencil radius, has grown to take in every row, from about $\delta = h/2$ on that node set; and the alternative of 30-node, degree-four stencils for the elastic rows themselves is stable through a resolved edge but violently unstable through a sharp one, as a degree-four basis across a jump is, which is presumably why the original work settled on 19 nodes and degree three. Annihilating the seed space is a choice, not a derivation: the seeds' true sixth derivatives at the anchor are not zero in a varying medium, and using them as right-hand sides is the alternative Section 7 records as future work.

Checks. Without contrast the seed weights equal the degree-three RBF-FD weights and the no-contrast weights of the jump path to 10^{-10} ; as $\delta \rightarrow 0$ they tend to the interface-aware weights at first order in δ on a real stencil, all four blocks, whichever side carries the standard monomials; every block is exact on the seed space through a $\delta = h/4$ edge and the Δ^3 blocks annihilate it; and the rebuilt rows of the assembled operator are exactly the stencils that see the edge, with no other entry moved. The jump path of Section 2.3 is unchanged bit for bit.

An ablation. To measure what the tangential structure of the seeds contributes, Section 6 uses a variant in which only the seeds of the pure normal monomials y^b are marched through the edge and every $x^a y^b$ with $a \geq 1$ is the plain monomial with the anchor material's stress. The plain columns come from a second chain on the material frozen at the anchor, since a frozen column must see frozen lower seeds on its right-hand side or $x^2 y^2$ would still be driven by the marched y^2 . The span has the same dimension and the same jets.

Cost. A 19-node march costs 21 to 25 ms and a seed row, with its 30-node march, 53 ms, in the reference implementation. On 10,000 nodes the operator through $\delta = h/8$ has 1965 seed rows and builds in 105 s; through $\delta = 0.01 = h$ it has 6719 and builds in 331 s. The marches run on a process pool and the assembled operators are cached, which is what the sweeps of Section 6 needed. Two optimisations were not built: the 19-node stencil is a prefix of the 30-node one, so one march with the seeds rescaled to the smaller $r_{\max}$ would serve both rows; and on a flat edge every stencil of one fixed row shares y_e and the profile, so one march with dense output could serve a whole row.

5.4. Curved edges. For a curved edge the material no longer depends on the normal coordinate of one frame, and three routes were open at the design

stage: (a) march the straight-edge seeds along the true normal through the stencil’s foot point and treat the curvature as a perturbation, $\kappa h \ll 1$ over a stencil; (b) solve (17) as a small boundary-value problem on a patch around each stencil, on a fine local grid with the harmonic polynomial as boundary data, the oversampling construction of multiscale finite elements [HW97] and the honest “one PDE per stencil”; or (c) compute harmonic coordinates once globally [OZ07] and take polynomials in them as the degree-one seeds. Route (a) is what was built, at its zeroth order.

The medium. With a curved interface the blend of (6) is taken in the true signed normal distance: the foot point comes from the same fixed-point iteration that places the stencil frames, and the distance is the displacement’s component along the normal there, so the foot point’s error enters squared and the distance is exact to rounding wherever the iteration converges at all. The two edges of the band are read on one periodic image of it, the image whose centre line is nearest; reading each edge on its own nearest image pairs the lower edge of one band with the upper edge of the next around $y = 0.75$, and was how the first version failed its own test.

The seeds. The profile of a rebuilt row carries the tangent angle θ at its foot point (x_0, y_0) and samples the medium along the true normal, $(x_0 - y \sin \theta, y_0 + y \cos \theta)$, with the march’s stops at the crossings of that line with each interface and its images, by Newton’s method from the vertical estimate. Along that line the blend is exactly the medium’s tanh profile in y for the interface the line is normal to (to 10^{-13}), so the seeds of a curved-edge stencil are the *flat* seeds of the same local coordinates (to 10^{-12} through a $\delta = 0.005$ edge on a stencil tilted by 0.08 radians): the ansatz extends the normal profile unchanged along the tangent. That is the locally flat approximation the jump stencils of Section 2.3 make, zeroth order in the curvature, and nothing else in the construction changes: the frames, the anchor at the evaluation node, the $r_{\max}$ scaling, the 19-node elastic rows and the 30-node Δ^3 rows are those of Section 5.3. A curved march costs $3.6 \times$ a flat one, 131 against 37 ms per 19-node stencil, because every material sample along the normal is a blend evaluation with two foot-point iterations and the march takes a few thousand of them.

What route (a) gets wrong. In the tangent frame a node at tangential offset x from the foot point lies at true normal distance $y - \kappa x^2/2 + O(\kappa^2)$ from the curve, while its seed value assumes y . Over a stencil radius $r \approx 2.5h$ that is $\kappa r^2/2$: for the sine interfaces of amplitude 0.02, whose curvature reaches $\kappa = 0.02(2\pi)^2 = 0.79$, it is 1.0×10^{-3} at 2500 nodes and 1.3×10^{-4} at 19,600, to be compared with δ : a tenth of a $\delta = 0.01$ edge at the coarse end and 5% of a $\delta = 0.0025$ edge at the fine end. So the geometric error of route (a) is largest where the seeds matter most, through the sharpest edge, and it shrinks with n only as h^2 . Section 6 measures what it costs on the node sets we can afford, and finds route (a) at the seed operator’s own floor on all of them, so neither (b) nor (c) was built. If a sharper edge or a tighter curve

ever asks for more, the cheaper first step is not the boundary-value problem but the curvilinear evaluation of the same marched seeds: evaluate each seed at the node's true arclength and normal distance instead of its tangent coordinates, and scale the tangential jets at the anchor by $1/(1 - \kappa y_e)$. That makes the material exact at every node at no march cost and leaves only the operator's curvature terms as the error; Section 7 lists it first among the things to do next.

6. TWO-DIMENSIONAL RESULTS

This section is the two-dimensional test of the construction of Section 5, on the flat and curved geometries of Section 2.2. It answers three questions in turn. Whether the seed-augmented operator is stable at the hyperviscosity the standard scheme was tuned for (Section 6.3); what the seeds do through a flat edge at normal incidence, which settles the rule of when to seed (Section 6.4); and whether the same holds at oblique incidence, where the tangential structure of the seeds is put to work (Section 6.5), and through a curved edge, where the seeds are marched along the true normal (Section 6.6). Section 6.1 describes the test problems and the reference solutions, and Section 6.2 what the standard scheme does on its own, which is what the seeds have to improve on.

6.1. Test problems and references. *Pulses and floors.* Every run in this section uses the medium, node sets, stencils, hyperviscosity and time step of Section 2.2, the node counts $n = 2500, 4900, 10,000$ and $19,600$ (h from 0.02 to 0.0071), and $t = 1$ as the measurement time. The pulse at normal incidence is the Gaussian plane P-wave of sharpness 15 started at $y = 0.875$, whose tails are 2×10^{-14} at the edges. We tried the dissertation's narrower pulse first (sharpness 23 from $y = 0.75$) and found its resolution floor, 10^{-2} in v on the finest node set against 10^{-7} in the one-dimensional study, high enough to hide everything but the jump; the wider pulse lowers the floor by $8\times$ at $19,600$ nodes, and the pulse can be widened no further if its tails are to clear both edges. Two floors are used. The *resolution floor* is the same pulse on the same nodes in the uniform background, measured against its exact translate: the error of the standard scheme with no edge at all. The *seed floor* is the seed operator's own: the same rows that a given δ would seed, built through a contrast of 10^{-6} , a medium the exact translated pulse cannot tell from uniform and through which the seeds equal the monomials to rounding, measured against the same translate. It exists because the seeded rows are a different scheme from the bulk, 19-node degree-three elastic rows with 30-node annihilating Δ^3 rows against 30-node degree-four stencils, with a resolution floor of their own.

The two operators. Both operators are run at every (n, δ) : the standard scheme, which samples the coefficients at the nodes, and the seed operator of Section 5.3, which at $\delta = 0$ is the interface-aware operator of Section 2.3 and for $\delta > 0$ puts the seed stencils on every row whose 19-node stencil sees

the edge, tails included. No rule about which edges to seed was imposed beforehand; the flat sweep is what sets the rule. On the flat geometry the seeded rows number 985, 1627, 2735 and 4888 at $\delta = 0.0025$ (39% down to 25% of the nodes, the 19δ tails plus a stencil radius), 1796, 3388, 6719 and 12,929 at $\delta = 0.01$ (72% to 66%), and every row at $\delta = 0.04$. Rates are measured in $h = n^{-1/2}$ between successive node sets. The tables give two significant digits, three when the leading digit is 1; the ratios quoted in the text are computed from the unrounded errors.

References. Each experiment has a reference that shares nothing with the stencils under test. For a jump on a flat interface it is the exact solution, the ray sum of Section 4 applied to the P wave. For a flat edge of width $\delta > 0$ at normal incidence the solution depends on y alone, and the reference is the one-dimensional pseudo-spectral solver of Section 4 on the image of the problem under $x = 1 - 2y$ (wave speed $2c_p$, density $\rho/2$, impedances unchanged), fed the exact image of the two-dimensional initial state, with σ_{xx} recovered exactly from σ_{yy} through $\partial_t \sigma_{xx} = \lambda v_y$. The two-dimensional blend of (6), linear in the weight for λ , μ and ρ , is mapped pointwise rather than rebuilt as a one-dimensional layer, since the one-dimensional profile (3) blends c and ρ and the two agree only in the jump limit. This reference tends to the ray sum at first order in δ (the largest difference in v at $t = 0.3$ is 0.167, 0.090, 0.046, 0.023 and 0.012 for $\delta = 0.016$ down to 0.001), doubling its grid or halving its step moves v , σ_{xx} and σ_{yy} by under 3×10^{-10} at $t = 1$ ($\delta = 0.005$), and it is run on 4096 nodes for $\delta = 0.0025$ and 1024 otherwise with $\Delta t = 5 \times 10^{-5}$. Because it depends on y only it is evaluated at every node directly, and no resampling enters the flat errors.

At oblique incidence on a flat medium the coefficients still depend on y only, so the Fourier modes of the state in x never mix: each mode is a one-dimensional system in y with $\partial_x \rightarrow 2\pi i k$ and complex coefficients, Fourier pseudo-spectral in y , and all modes march together under RK4. The reference is exact in x (13 modes carry the train of Section 6.5 to 10^{-14}), uses 4096 points in y for $\delta = 0.0025$, and costs minutes (220 s at $\delta = 0.0025$, 40 s at 0.01). Its checks: the train in a uniform medium is its exact translate to 10^{-10} and has zero curl; at normal incidence the solver agrees with the mapped one-dimensional reference to 10^{-10} , two independent code paths; through an edge at oblique incidence the elastic energy is conserved to 10^{-12} , and doubling the grid while halving the step moves the answer by 10^{-8} . The fields and their derivatives are evaluated at the nodes by trigonometric interpolation in both directions.

For a curved edge of width $\delta > 0$ the modes mix, and the reference is the same solver on an $n_x \times n_y$ product grid: derivatives by FFT along each axis, the material multiplied in physical space, and RK4 with the step $\Delta t = 0.5/(c_{p,\max} \sqrt{n_x^2 + n_y^2})$, the CFL number 0.5 on the grid's largest wavenumber. The grids are 512×1024 for $\delta = 0.01$, 1024×2048 for 0.005 and 2048×4096 for 0.0025. On a flat medium at oblique incidence the

product-grid solver agrees with the Fourier-in- x one to 10^{-11} , two representations of one scheme; on a curved edge the energy is conserved to 10^{-9} , and the curl is no longer zero, because a curved edge converts P to S. Its error bars, measured at $t = 1$ on 6000 random points: at $\delta = 0.01$ the 512×1024 grid moves by 1.3×10^{-9} when n_x is halved and by 3.1×10^{-9} when both directions are doubled or the step is halved (the same number, because the doubled grid's difference is its halved step), and at $\delta = 0.005$ the 1024×2048 grid moves by 8×10^{-11} and 2.6×10^{-10} under doubling in x and in y ; both are five orders below anything measured. A jump on a curved interface has no product-grid reference, and for it we use the interface-aware operator of Section 2.3 on a node set of its own, 122,500 nodes ($h = 1/350$), resampled onto the sweep's nodes by one-sided interpolation that never crosses an interface. Its bar is about 5% of the finest point: runs of 62,500 and 122,500 nodes differ by 1.7×10^{-4} in v on the 19,600-node set, where the 19,600-node interface-aware run's error is 1.39×10^{-3} against the finer reference and 1.25×10^{-3} against the coarser, a reference made by the same scheme flattering a run of it by the correlation of their errors; the finer one's numbers are the ones reported. The references cost 201 s (512×1024), 1952 s (1024×2048), 3.6 h (2048×4096) and 286 s (the jump run) to $t = 1$, and are cached.

Errors. As fixed in Section 2.2: relative ℓ^2 errors in v and in σ_{yy} at $t = 1$ over all nodes, and for u the largest $|u|$ at normal incidence on a flat edge, where the true u is identically zero, and the relative ℓ^2 error in u at oblique incidence and on the curved geometry, where it is not (a curved edge makes u of size 0.12 out of the incident pulse at normal incidence, and its relative error then measures the scattered field alone).

6.2. Naive baseline. The naive rows of Tables 3 and 4 are the standard scheme through flat edges of width $\delta = 0, 0.0025, 0.01$ and 0.04 at normal incidence, and the floor rows are the same pulse with no edge. In the ratio h/δ the three widths run from 8 to 2.9 (never resolved), from 2 to 0.71 (resolved from 10,000 nodes on) and from 0.5 to 0.18 (resolved on every node set), the three regimes of Section 4. Four things are to be read off before the seeds enter.

The resolution floor is the first. Through the edge every node set resolves, $\delta = 0.04$, the naive error in v is the floor to within 7% at every n , with the floor's rates: a resolved edge costs the standard scheme nothing. Through the edge no node set resolves, $\delta = 0.0025$, the naive scheme behaves like the jump: second order in v (rates 1.9, 1.6, 2.2), $10\times$ the floor at 19,600 nodes, and at 0.6 to 0.75 of the jump's error, since the rows that straddle the edge centre at $h/2$ see $\tanh(h/2\delta)$ of the contrast, 0.9993 at 2500 nodes and 0.89 at 19,600. The knee is there, at $\delta = 0.01$, but small in v : the naive error exceeds the floor by 19%, 17%, 18% and 7% from $h = 2\delta$ to $h = 0.71\delta$, so that the edge's own contribution, the excess in quadrature, falls from 3.3×10^{-2} to 5.2×10^{-4} , faster than second order and accelerating,

TABLE 3. Flat edges at normal incidence: relative ℓ^2 error in v at $t = 1$ and the rates between successive node sets, for the standard scheme (naive) and the seed operator (the interface-aware stencils at $\delta = 0$), with the resolution floor. h/δ runs from 8 to 2.9 at $\delta = 0.0025$, from 2 to 0.71 at 0.01 and from 0.5 to 0.18 at 0.04.

δ	operator	$n = 2500$	4900	10000	19600	rates
0 (jump)	naive	1.26×10^{-1}	7.2×10^{-2}	3.6×10^{-2}	2.2×10^{-2}	1.7, 2.0, 1.5
0 (jump)	interface-aware	5.4×10^{-2}	1.82×10^{-2}	4.7×10^{-3}	1.23×10^{-3}	3.2, 3.8, 4.0
0.0025	naive	8.9×10^{-2}	4.8×10^{-2}	2.7×10^{-2}	1.29×10^{-2}	1.9, 1.6, 2.2
0.0025	seeds	3.0×10^{-2}	7.7×10^{-3}	2.1×10^{-3}	6.4×10^{-4}	4.1, 3.7, 3.5
0.01	naive	6.2×10^{-2}	2.1×10^{-2}	5.7×10^{-3}	1.39×10^{-3}	3.1, 3.7, 4.2
0.01	seeds	3.0×10^{-2}	1.05×10^{-2}	3.9×10^{-3}	1.57×10^{-3}	3.1, 2.8, 2.7
0.04	naive	5.2×10^{-2}	1.94×10^{-2}	5.1×10^{-3}	1.37×10^{-3}	2.9, 3.7, 3.9
0.04	seeds	7.5×10^{-2}	2.8×10^{-2}	7.7×10^{-3}	2.7×10^{-3}	2.9, 3.6, 3.1
floor	naive, uniform	5.2×10^{-2}	1.83×10^{-2}	4.8×10^{-3}	1.29×10^{-3}	3.1, 3.7, 3.9

which is the knee shape of Section 4 seen through a high floor. And the spurious u separates the cases an order of magnitude more sharply than v does. The naive stencils excite u where a field varies sharply in y , because ∂_x on scattered nodes does not annihilate such a field exactly, and u carries none of the pulse’s own dispersion error, so its floor is low (8.9×10^{-4} down to 3.0×10^{-5}). Against that floor the jump is $11\times$, $24\times$, $46\times$ and $194\times$ from 2500 to 19,600 nodes and $\delta = 0.0025$ is $11\times$, $24\times$, $39\times$ and $116\times$: an unresolved edge’s error decays like the jump’s, slower than the floor, and the ratio grows with n . The resolved edge equals the floor at every n . The edge of width 0.01 sits $3\times$, $3\times$, $6\times$ and $3\times$ above it, rising while $h \geq \delta$ and falling back once $h < \delta$, a drop of $9\times$ between 10,000 and 19,600 nodes against $4\times$ for the floor: the knee of Section 4, at $h = \delta$ as there, in the one quantity the floor does not pollute.

Two remarks. The two-dimensional naive scheme is not first order through an unresolved edge as one-dimensional FD4 was: the fixed rows straddling the edge centre keep the coefficient sampling symmetric, and RBF-FD through a jump is already second order with a large constant [MF17a], so the knee is a change of constant rather than of order and shows as a bend rather than a kink. And the seeds can at most remove the excess over the floor: a factor 10 in v and up to 116 in spurious u through $\delta = 0.0025$, 8 to 19% in v and 3 to $6\times$ in u through $\delta = 0.01$, nothing through $\delta = 0.04$. So the $\delta = 0.0025$ column and the spurious u are the primary evidence of the flat sweep, and a knee plot in v alone would show little.

6.3. Stability. The empirical stability recipe for interface-aware RBF-FD, Δ^3 hyperviscosity at $\gamma = 2.4 \times 10^{-11}(h/0.02)^5$ with the fixed rows straddling each interface, was found empirically [Mar16, MF17a], and the question of

TABLE 4. Flat edges at normal incidence: the largest spurious $|u|$ at $t = 1$ (the true u is identically zero) for the runs of Table 3.

δ	operator	$n = 2500$	4900	10000	19600
0 (jump)	naive	9.6×10^{-3}	1.09×10^{-2}	5.7×10^{-3}	5.7×10^{-3}
0 (jump)	interface-aware	1.24×10^{-3}	4.6×10^{-4}	1.50×10^{-4}	3.3×10^{-5}
0.0025	naive	9.5×10^{-3}	1.06×10^{-2}	4.7×10^{-3}	3.4×10^{-3}
0.0025	seeds	1.73×10^{-3}	6.5×10^{-4}	1.86×10^{-4}	5.1×10^{-5}
0.01	naive	2.7×10^{-3}	1.53×10^{-3}	7.6×10^{-4}	8.5×10^{-5}
0.01	seeds	3.5×10^{-3}	1.08×10^{-3}	4.3×10^{-4}	1.13×10^{-4}
0.04	naive	1.19×10^{-3}	4.8×10^{-4}	1.00×10^{-4}	2.4×10^{-5}
0.04	seeds	5.9×10^{-3}	1.75×10^{-3}	5.7×10^{-4}	1.25×10^{-4}
floor	naive, uniform	8.9×10^{-4}	4.5×10^{-4}	1.22×10^{-4}	3.0×10^{-5}

this section is whether it survives an augmentation that is no longer polynomial. It was measured on the 900-node flat set (4500 eigenvalues, dense) at the standard γ and CFL 0.5, by the largest real part of the spectrum with the hyperviscosity in and by the largest RK4 amplification $\max |R(\lambda \Delta t)|$ over it; 1.0004 is the naive scheme's own value on that node set, a mode growing by 4×10^{-4} per step and harmless over a run, and anything above 1.001 is a run that visibly grows. Table 5 collects the operators tried.

What fails. The first construction gave each 19-node seed row its own Δ^3 row on the same 19 nodes with the seeds annihilated, the exact twin of the jump path. It is fine while the seed rows form a band around each edge ($+6.2 \times 10^{-2}$ at $\delta = h/8$, $+8.7 \times 10^{-2}$ at $h/4$, the naive scheme's level) and unstable once the tails, 19δ plus the stencil radius, make every row a seed row, from about $\delta = h/2$ on this node set: $+1.05$ at $h/2$ and $+2.65$ at h , with $\max |R|$ of 1.003 and 1.018. The naive spectrum does not move with δ , so the movement is the seed rows', but it is not the seeds. Plain 19-node degree-three stencils everywhere on the jump medium, with no seeds at all, give $+2.4$ and worse: the 19-node degree-three scheme is unstable at the γ tuned for the 30-node degree-four one, and the jump path never sees this because its band of rebuilt rows is $4h$ wide. The reason is at the damping end of the spectrum: the extreme eigenvalue of the Δ^3 block of the all-seed operator is -82 against -151 for the naive one, half the damping, because a 19-node stencil carrying 20 coupled constraints leaves 18 degrees of freedom for the Gaussian part of a sixth-order operator and what comes out is barely a Δ^3 . Trimming the seed rows with a tolerance on the material spread does not help (at $\delta = h/2$, 574, 458 and 420 rows in place of 900 leave $+1.05$, $+1.05$ and $+0.35$): it is the width of the contiguous 19-node region that matters, not the count. Doubling γ , everywhere or on the seed rows alone, stabilises every width ($+2.9 \times 10^{-2}$ at $h/2$, $+3.9 \times 10^{-2}$ at h , and 1.5γ does as well) at the price of the extra damping. Building the seed rows themselves on the naive footprint, 30 nodes and degree four, is stable through a resolved edge

(+0.10, +0.05, +0.06 at $\delta = h/2, h, 1.9h$) and violently unstable through a sharp one (+30 and +28 at $h/8$ and $h/4$, $\max |R|$ up to 1.23), exactly as the jump path is with degree-four stencils across the interface (+17, $\max |R|$ 1.12): a degree-four basis across a sharp feature is the problem, presumably why the original work settled on 19 nodes and degree three, and conditioning is not the cause (the 30-node seed blocks are at 85 to 215). Keeping the naive Δ^3 rows on the seed rows is stable everywhere, but the naive Δ^3 does not annihilate the seeds, and the seeds are what the scheme resolves through the edge.

What works. The elastic rows keep the 19-node seed stencils, and the Δ^3 rows of the same nodes are built on the naive 30-node footprint with the same seeds annihilated, the construction of Section 5.3. It is stable at every width at the standard γ , $+4.1 \times 10^{-2}$ to $+6.6 \times 10^{-2}$ with $\max |R| \leq 1.0005$, and the extreme Δ^3 eigenvalue is back at -176 : the naive damping restored on the seed rows. As a check that the choice is not bought with accuracy, the variants were run on 2500 nodes through $\delta = h/8$ with the pulse of Section 6.1: every seed variant removes most of the naive excess (spurious u down 4 to $9\times$, to 1 to $2.5\times$ the floor, and v to the floor), the chosen one is the best in v (3.0×10^{-2} against 4.4×10^{-2} with the naive Δ^3 rows kept and 4.7×10^{-2} with 19-node rows, below the naive floor of 5.2×10^{-2}) and keeps the most energy ($E(1)/E(0) = 0.995$), and the 2γ variant buys the smallest u with twice the chosen operator's error in v . The answer to the question is therefore yes: RBF-FD stability survives seed-augmented stencils at the standard γ and with the straddling rows kept, provided the hyperviscosity row of a seed stencil has the footprint γ was tuned for. Annihilating the seed space was the only variant tried that both annihilates and is stable, and the alternative of Section 5.3, the seeds' true sixth derivatives as right-hand sides, was never needed.

Acceptance on 2500 nodes. Table 6 and Figure 6 are the chosen operator on the 2500-node set (12,500 eigenvalues per operator) at $\delta = h/8, h/2$ and $2h$, with the pulse run to $t = 1$ against the references of Section 6.1. The seed operator's rightmost eigenvalues sit at $\max |R| = 1.0003$ to 1.0004 whatever δ , the level of the jump-aware operator (1.0002) and of the naive scheme on 900 nodes (1.0004), while the naive scheme on 2500 nodes is at 1.00001: the rebuilt rows carry a slightly larger rightmost eigenvalue than plain stencils, as the interface rows of the jump case do, and no more. The energy ratios say the same in the time domain, 0.995 at $\delta = h/8$ and 1.003 at $h/2$ against a quadrature drift of $\pm 5\%$ for the exact solution on these nodes (the floor loses 7.5%). At $\delta = 2h$, where every row is a seed row, the ratio is 1.012 and keeps rising when the run is continued, 1.045, 1.055, 1.063 and 1.071 at $t = 1.5, 2, 2.5$ and 3, about 0.012 per unit time, while the naive scheme (0.999 at $t = 3$), the jump-aware operator (0.96) and the seed operator at $\delta = h/8$ (1.006) stay bounded: a weakly excited mode near the +0.1 rightmost eigenvalue, far below what $\max |R| = 1.0004$ would allow

TABLE 5. The operators tried on the 900-node flat set, at the standard γ and CFL 0.5: rebuilt rows, the largest real part of the spectrum with the hyperviscosity in, and the largest RK4 amplification over it. “Seeds” is the operator of Section 5.3; the four variants below it at each width are the rejected constructions of the text; the last row at each width is the control with plain 19-node degree-three stencils everywhere and no seeds.

δ	operator	rebuilt rows	max Re λ , with γ	RK4 max $ R $
jump	naive	0	$+5.9 \times 10^{-2}$	1.00039
jump	interface-aware	460	$+4.2 \times 10^{-2}$	1.00029
$h/8$	naive	0	$+5.9 \times 10^{-2}$	1.00039
$h/8$	seeds	513	$+4.1 \times 10^{-2}$	1.00028
$h/8$	seeds, 19-node Δ^3 rows	513	$+6.2 \times 10^{-2}$	1.00042
$h/8$	seeds, naive Δ^3 rows kept	513	$+9.1 \times 10^{-2}$	1.00062
$h/8$	seeds, 30-node degree-4 rows	555	$+3.1 \times 10^1$	1.23066
$h/8$	19-node degree-3 rows, no seeds	0	$+2.4 \times 10^0$	1.01166
$h/4$	naive	0	$+5.8 \times 10^{-2}$	1.00039
$h/4$	seeds	634	$+5.4 \times 10^{-2}$	1.00036
$h/4$	seeds, 19-node Δ^3 rows	634	$+8.7 \times 10^{-2}$	1.00059
$h/4$	seeds, naive Δ^3 rows kept	634	$+7.8 \times 10^{-2}$	1.00053
$h/4$	seeds, 30-node degree-4 rows	676	$+2.8 \times 10^1$	1.21168
$h/4$	19-node degree-3 rows, no seeds	0	$+2.6 \times 10^0$	1.01286
$h/2$	naive	0	$+1.48 \times 10^{-2}$	1.00010
$h/2$	seeds	900	$+6.6 \times 10^{-2}$	1.00045
$h/2$	seeds, 19-node Δ^3 rows	900	$+1.05 \times 10^0$	1.00293
$h/2$	seeds, naive Δ^3 rows kept	900	$+2.8 \times 10^{-2}$	1.00019
$h/2$	seeds, 30-node degree-4 rows	900	$+1.04 \times 10^{-1}$	1.00071
$h/2$	19-node degree-3 rows, no seeds	0	$+3.3 \times 10^0$	1.01793
h	naive	0	$+2.0 \times 10^{-3}$	1.00001
h	seeds	900	$+5.2 \times 10^{-2}$	1.00035
h	seeds, 19-node Δ^3 rows	900	$+2.6 \times 10^0$	1.01818
h	seeds, naive Δ^3 rows kept	900	$+9.3 \times 10^{-3}$	1.00006
h	seeds, 30-node degree-4 rows	900	$+5.3 \times 10^{-2}$	1.00036
h	19-node degree-3 rows, no seeds	0	$+3.7 \times 10^0$	1.02042
$1.9h$	naive	0	$+1.08 \times 10^{-4}$	1.00000
$1.9h$	seeds	900	$+5.6 \times 10^{-2}$	1.00038
$1.9h$	seeds, 19-node Δ^3 rows	900	$+9.3 \times 10^{-1}$	1.00262
$1.9h$	seeds, naive Δ^3 rows kept	900	$+4.2 \times 10^{-3}$	1.00003
$1.9h$	seeds, 30-node degree-4 rows	900	$+5.7 \times 10^{-2}$	1.00039
$1.9h$	19-node degree-3 rows, no seeds	0	$+3.0 \times 10^0$	1.01655

(10% per unit time), confined to the regime in which every row is a seed row, which is the regime the rule of Section 6.4 excludes. The same operator on the curved geometry is in Section 6.6, row for row the same.

6.4. The flat sweep and the rule. Tables 3, 4 and 8 and Figure 7 are the two-dimensional twin of the knee experiment of Section 4: both operators

TABLE 6. The chosen operator on the 2500-node flat set at three edge widths, against the naive scheme, the interface-aware operator at the jump and the uniform-medium floor: rebuilt rows, the spectrum's largest real part with the hyper-viscosity in, the largest RK4 amplification, the energy ratio $E(1)/E(0)$ of the run, its largest spurious $|u|$ and its error in v .

δ	operator	rebuilt rows	max Re λ , with γ	RK4 max $ R $	$E(1)/E(0)$	max $ u $	error in v
jump	naive	0	$+3.5 \times 10^{-3}$	1.00001	0.969	9.6×10^{-3}	1.26×10^{-1}
jump	interface-aware	805	$+4.9 \times 10^{-2}$	1.00020	0.979	1.24×10^{-3}	5.4×10^{-2}
(floor)	naive, uniform medium	0	$+3.5 \times 10^{-14}$	1.00000	0.925	8.9×10^{-4}	5.2×10^{-2}
$h/8$	naive	0	$+3.5 \times 10^{-3}$	1.00001	0.969	9.5×10^{-3}	8.9×10^{-2}
$h/8$	seeds	985	$+6.3 \times 10^{-2}$	1.00026	0.995	1.73×10^{-3}	3.0×10^{-2}
$h/2$	naive	0	$+4.6 \times 10^{-3}$	1.00002	0.992	2.7×10^{-3}	6.2×10^{-2}
$h/2$	seeds	1796	$+7.8 \times 10^{-2}$	1.00032	1.003	3.5×10^{-3}	3.0×10^{-2}
$2h$	naive	0	$+6.7 \times 10^{-5}$	1.00000	0.988	1.19×10^{-3}	5.2×10^{-2}
$2h$	seeds	2500	$+9.6 \times 10^{-2}$	1.00039	1.012	5.9×10^{-3}	7.5×10^{-2}

at every (n, δ) through the flat edges at normal incidence, with the seeds on every row that sees the edge.

An edge the nodes never resolve. Through $\delta = 0.0025$, h/δ from 8 to 2.9, the seeds are fourth order at every resolution, rates 4.1, 3.7, 3.5 in v and 3.6, 3.2, 3.5 in σ_{yy} against 1.6 to 2.2 for the naive scheme, which is $3.0\times$, $6.2\times$, $13\times$ and $20\times$ worse in v and $5.5\times$, $16\times$, $25\times$ and $67\times$ worse in spurious u from 2500 to 19,600 nodes; the seeds' u sits at 1.5 to $1.9\times$ the floor's, where the naive scheme's sits at 11 to $116\times$. That is the one-dimensional result in two dimensions: the order of the jump-aware stencils, reached without the nodes seeing the edge, on the same nodes and the same time step. The constant is not the jump row's but about half of it at every n (3.0×10^{-2} against 5.4×10^{-2} at 2500 nodes, 6.4×10^{-4} against 1.23×10^{-3} at 19,600), and half the uniform-medium floor's. That is the seed operator's own floor, not the edge: seeding the same rows through a contrast of 10^{-6} gives 2.7×10^{-2} , 6.2×10^{-3} , 1.9×10^{-3} and 6.9×10^{-4} , half the naive floor at every n and within 10 to 25% of the seed errors through $\delta = 0.0025$. So the seeds remove the edge error down to their own resolution floor, as the jump-aware operator removes it down to the naive one. Why a band of 19-node degree-three seed rows with 30-node annihilating Δ^3 rows is more accurate on this pulse than 30-node degree-four stencils everywhere is not settled. It is not the degree, since plain 19-node degree-three stencils everywhere with their own Δ^3 rows are 2 to $4\times$ worse than the standard scheme (9.5×10^{-2} down to 4.9×10^{-3} ; the operator Section 6.3 found unstable at the standard γ , harmless to $t = 1$), and it is not less damping, since the energy ratios match the floor's (0.93, 0.92, 0.99, 0.99 for both). The variants of Section 6.3 point at the Δ^3 rows, and a dispersion study of those rows in a uniform medium would settle it; nothing in what follows depends on it.

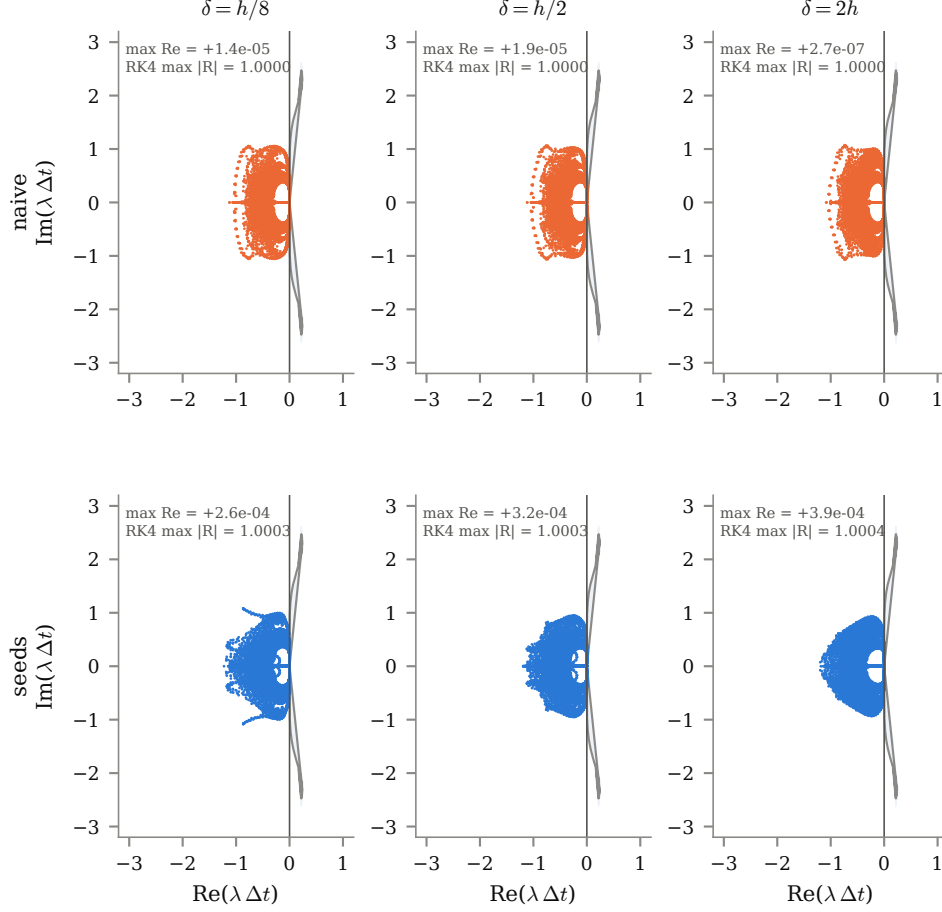


FIGURE 6. Spectra of the naive (top) and seed (bottom) operators on the 2500-node flat set through edges of width $\delta = h/8$, $h/2$ and $2h$, scaled by the time step, with the RK4 stability boundary. The annotations give the largest real part of $\lambda \Delta t$ and the largest amplification over the spectrum.

The crossover. Through $\delta = 0.01$ the seeds lead by $2.1\times$ in v at $h = 2\delta$ (with u equal), $2.0\times$ at $h = 1.4\delta$ (u $1.4\times$) and $1.5\times$ at $h = \delta$ (u $1.8\times$), and trail by $0.88\times$ (u $0.75\times$) at $h = 0.71\delta$. Across the panel the seed rates fall from 3.1 to 2.7 while the naive rates rise from 3.1 to 4.2 : the naive scheme is resolving the edge, and the seeds are seeding rows that no longer need it, 66 to 72% of the nodes here. Through the edge every node set resolves, $\delta = 0.04$, where every row is a seed row, the seeds are 1.4 to $2.0\times$ worse than the naive scheme in v and 3.7 to $5.6\times$ worse in u , at rates of 2.9 to 3.6 against 2.9 to 3.9 , as the 2500-node acceptance run already found. The naive scheme is at its floor to within 7% at every n ; there is nothing left for the

seeds to remove, and the 19-node degree-three stencils through a resolved profile are the less accurate scheme. The break-even is between $h = \delta$ and $h = 0.71\delta$, and the estimate $\delta \lesssim h/4$ that the three points of Section 6.3 suggested was too cautious: at $\delta = h/2$ the seeds halve v and only just lose in u .

The rule. Seed when $\delta \leq h$, and run the standard scheme otherwise. The rule is per edge, δ the edge's width and h the node set's spacing, not per row: the tolerance on the material spread of Section 5.3 trims the tails of the seeded region, and the accuracy lost on a resolved edge comes from the rows at its centre, not its tails. With the rule the seed operator is never worse than the naive scheme and is at or below the resolution floor wherever it is used; in this sweep it means every node set at $\delta = 0.0025$, up to 10,000 nodes at $\delta = 0.01$ and never at $\delta = 0.04$. The comparison is against coefficient sampling on the same nodes; the coefficient treatments of seismic finite differences were run on this operator too (below) and on the one-dimensional scheme (Section 4), and none of them reaches the seeds' order. The rule applies in one dimension as well (Section 3.5), where the resolved edge $\delta = 0.04$ of Table 1 is the case in which the two schemes agree to three digits and nothing is gained by seeding.

The standing alternative, on the scattered nodes. The coefficient treatments of Section 4 were run on the naive operator of this section as well (Table 7): the wave moduli $K = \lambda + 2\mu$ and μ averaged harmonically and the density arithmetically over a square of side h or $2h$ about each node, isotropy kept; the same three fields band-limited with the separable windowed sinc of [KWR22]; and the edge widened to $\max(\delta, h)$ and $\max(\delta, 2h)$; each against the true medium's reference, on the flat geometry through $\delta = 0.0025$ and 0.01 and the curved one through 0.0025 , 0.005 and 0.01 . On scattered nodes no node sits at a fixed offset from the edge, so the one-cell mean acts everywhere. Through $\delta = 0.0025$, which no node set resolves, it is the best treatment on either geometry: level with sampling at 2,500 nodes and 2.1 to $3.6\times$ below it at 10,000 and 19,600, with rates 3.4 to 3.6 over the last two doublings, and the seeds of Tables 3 and 12 are 3.1 to $5.9\times$ below it at every n . The band-limited coefficients are level with or above sampling on the two coarse sets and 1.5 to $1.7\times$ below it on the two fine ones; the two-cell mean, the best treatment in one dimension, is 1.05 to $1.4\times$ above sampling at every n here, second order throughout. Where the knee sits inside the sweep the treatments cross above sampling as in one dimension, and earlier for the wider ones: through $\delta = 0.01$ the one-cell mean is $1.1\times$ below sampling while $h \geq 1.4\delta$ and $2.4\times$ above it at $h = 0.7\delta$, the band-limited coefficients 1.2 to $1.9\times$ above at every n , the two-cell mean 1.8 to $10\times$; through the curved $\delta = 0.005$ the one-cell mean's crossover lies between $h = 2\delta$ and 1.4δ , before the seeds' at $h \approx \delta$. The widened edge is 5 to $120\times$ worse than sampling wherever it acts, on both geometries: the exact solutions through edges of width 0.02 and 0.04 differ from the true $\delta = 0.01$ one by 27% and

82% in v at $t = 1$, so a medium changed over a cell or two costs far more here than in one dimension, where the pulse crosses the layer once; here it crosses the band twice by $t = 1$ and converts at each of the four crossings. The spurious u tells the same story: through $\delta = 0.0025$ the one-cell mean cuts it by 1.3 to $1.8\times$, the seeds by 5.5 to $67\times$. The anisotropic effective medium of Schoenberg and Muir, which [KWR22] find best in elastic media, needs an anisotropic five-field operator the implementation does not have and was not built; Section 7 lists it. As in one dimension, this is one operator, one contrast and one pulse, with our isotropic reading of treatments their sources run on staggered Cartesian grids.

The naive knee, and the stress. In v the naive knee is the 8 to 19% excess over the floor of Section 6.2; in spurious u it is the plateau at 3 to $6\times$ the floor while $h \geq \delta$ and the drop of $9\times$ between 10,000 and 19,600 nodes once $h < \delta$, visible in the $\delta = 0.01$ panel of Figure 7. The error in σ_{yy} (Table 8) says what v says: fourth order for the seeds through $\delta = 0.0025$, from 2.3×10^{-2} down to 6.7×10^{-4} , $18\times$ below the naive scheme at 19,600 nodes, and the same crossover at $\delta = 0.01$ and the same loss at 0.04.

The coarse-grid picture. Figure 8 is the run on 10,000 nodes through edges of width $\delta = 0.0025 = h/4$ at $t = 1$, the reference and the two error maps in v on one colour scale, resampled one-sided to a pixel grid. The naive error is 2.7% of the pulse with a spurious u of 4.7×10^{-3} ; the seeds' is 0.21% with 1.9×10^{-4} . At $t = 0.25$, when the pulse has just split at the upper edge, the numbers are 1.5% and 3.5×10^{-3} against 0.07% and 1.9×10^{-4} : the naive error is the horizontal streaking along the edge that the stencils straddling it produce, already spread over the whole pulse, and by $t = 1$ it has been carried everywhere the reflections went, while the seeds' 0.2% is the resolution error of the pulse itself.

6.5. Oblique incidence. At normal incidence on a flat edge the solution is independent of x and $u \equiv 0$, so the seeds of the monomials $x^a y^b$ with $a \geq 1$ can show only through the weights, never through the solution. This experiment sends the wave in at an angle, where $u \neq 0$, the tangential strain of the incident P wave meets the edge, and P-to-S conversion appears.

The train. A single tilted plane pulse cannot be periodic on the unit square and stay out of the band, so the initial data is the doubly periodic train of Section 2.2: crests $x - 2y = \text{const}$, at 26.6° to the edge normal, travelling towards $+x$ and $-y$, one crest through $(0, 0.875)$ repeated every 0.447 along the direction of travel with the Gaussian profile of sharpness 15, so that neighbouring crests overlap at 1.3×10^{-5} and 13 Fourier modes in x carry the train to 10^{-14} . The direction $(1, 2)$ has the largest crest spacing of any lattice direction below the critical angle (P-to-P transmission into the band turns critical at 45°); at 26.6° the transmitted P, transmitted S and reflected S waves leave at 39° , 21° and 15° , all propagating. A quarter of each strip starts inside the band. Every point carries the *background* material's P eigenvector, so that every field is one smooth profile everywhere and the

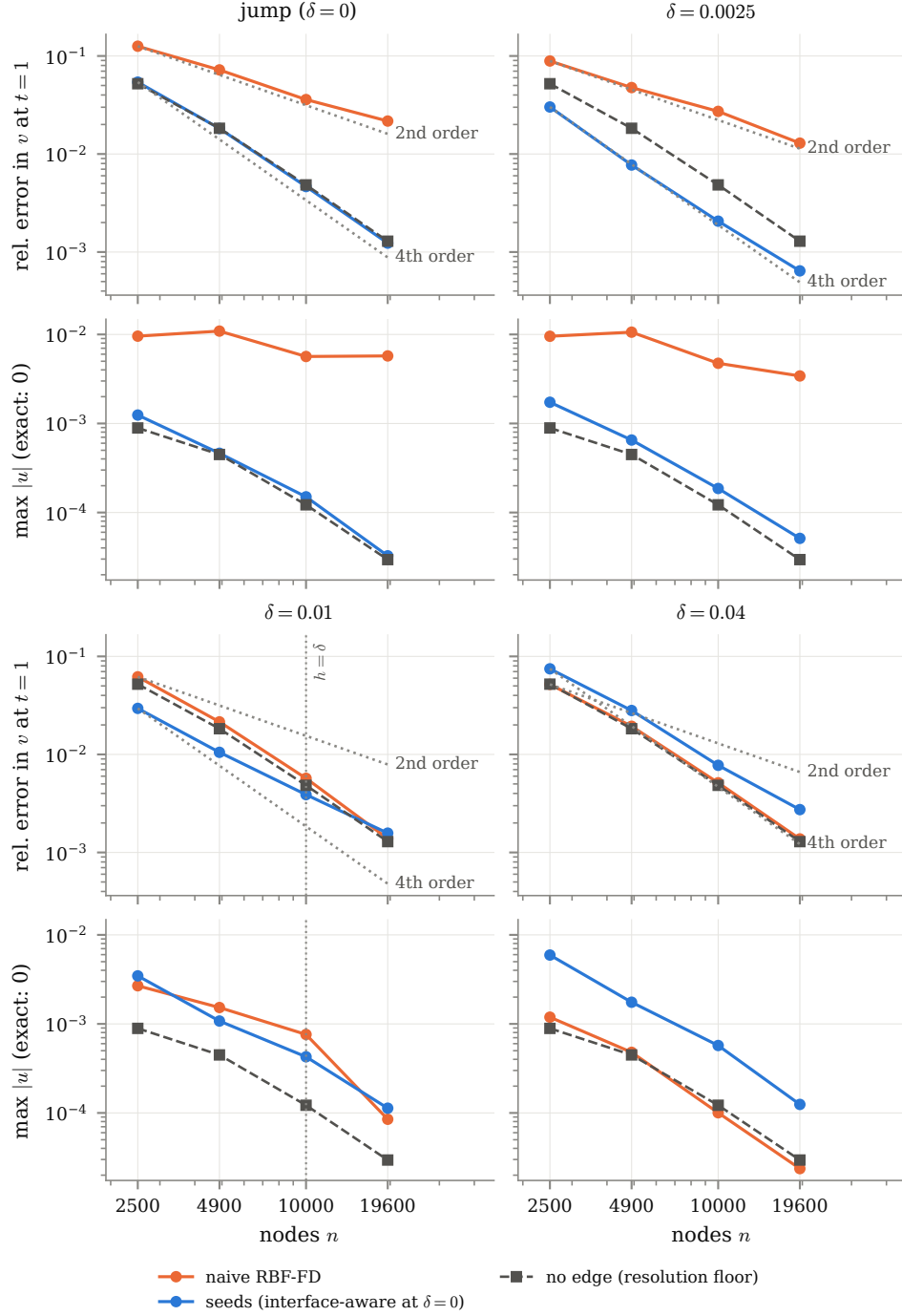


FIGURE 7. The flat sweep of Tables 3 and 4: relative error in v (upper panels) and largest spurious $|u|$ (lower panels) against n for the jump and three edge widths, naive scheme against seeds, with the uniform-medium floor and second- and fourth-order guides; the vertical line in the $\delta = 0.01$ panel is the node set at which $h = \delta$. Through $\delta = 0.0025$ the seeds lie on a fourth-order line below the floor; through $\delta = 0.04$ they lie above the naive scheme, which is at the floor.

TABLE 7. The naive operator on a treated medium against the naive operator on the sampled medium: relative ℓ^2 error in v at $t = 1$ against the true medium's reference on the flat geometry (top) and the curved one (bottom), the rates between successive node sets, and the resolution floor. The seeds on the same node sets are in Tables 3 and 12.

δ	operator	$n = 2500$	4900	10000	19600	rates
0.0025	naive	8.9×10^{-2}	4.8×10^{-2}	2.7×10^{-2}	1.29×10^{-2}	1.9, 1.6, 2.2
0.0025	widened, h	4.4×10^{-1}	3.0×10^{-1}	1.89×10^{-1}	1.19×10^{-1}	1.2, 1.2, 1.4
0.0025	widened, $2h$	9.1×10^{-1}	6.5×10^{-1}	4.3×10^{-1}	2.9×10^{-1}	1.0, 1.2, 1.2
0.0025	cell mean, h	9.3×10^{-2}	4.2×10^{-2}	1.22×10^{-2}	3.6×10^{-3}	2.4, 3.4, 3.6
0.0025	cell mean, $2h$	1.21×10^{-1}	6.1×10^{-2}	2.9×10^{-2}	1.35×10^{-2}	2.0, 2.1, 2.2
0.0025	band-limited	1.01×10^{-1}	4.9×10^{-2}	1.84×10^{-2}	7.6×10^{-3}	2.2, 2.7, 2.6
0.01	naive	6.2×10^{-2}	2.1×10^{-2}	5.7×10^{-3}	1.39×10^{-3}	3.1, 3.7, 4.2
0.01	widened, h	3.0×10^{-1}	1.25×10^{-1}	5.7×10^{-3}	1.39×10^{-3}	2.6, 8.7, 4.2
0.01	widened, $2h$	8.3×10^{-1}	5.4×10^{-1}	2.8×10^{-1}	1.14×10^{-1}	1.3, 1.8, 2.7
0.01	cell mean, h	5.9×10^{-2}	1.93×10^{-2}	6.5×10^{-3}	3.3×10^{-3}	3.3, 3.1, 2.0
0.01	cell mean, $2h$	1.14×10^{-1}	5.7×10^{-2}	2.8×10^{-2}	1.46×10^{-2}	2.1, 2.0, 1.9
0.01	band-limited	7.5×10^{-2}	2.6×10^{-2}	7.8×10^{-3}	2.5×10^{-3}	3.1, 3.4, 3.3
floor	naive, uniform	5.2×10^{-2}	1.83×10^{-2}	4.8×10^{-3}	1.29×10^{-3}	3.1, 3.7, 3.9
δ	operator	$n = 2500$	4900	10000	19600	rates
0.0025	naive	9.2×10^{-2}	4.6×10^{-2}	2.4×10^{-2}	1.19×10^{-2}	2.1, 1.9, 2.1
0.0025	widened, h	4.5×10^{-1}	2.9×10^{-1}	1.83×10^{-1}	1.15×10^{-1}	1.3, 1.3, 1.4
0.0025	widened, $2h$	9.3×10^{-1}	6.5×10^{-1}	4.3×10^{-1}	2.8×10^{-1}	1.0, 1.2, 1.2
0.0025	cell mean, h	9.0×10^{-2}	3.8×10^{-2}	1.12×10^{-2}	3.4×10^{-3}	2.6, 3.4, 3.6
0.0025	cell mean, $2h$	1.22×10^{-1}	6.0×10^{-2}	2.7×10^{-2}	1.27×10^{-2}	2.1, 2.2, 2.2
0.0025	band-limited	9.7×10^{-2}	4.5×10^{-2}	1.64×10^{-2}	6.9×10^{-3}	2.3, 2.8, 2.6
0.005	naive	8.1×10^{-2}	3.2×10^{-2}	6.3×10^{-3}	1.90×10^{-3}	2.8, 4.5, 3.5
0.005	widened, h	4.0×10^{-1}	2.4×10^{-1}	1.25×10^{-1}	5.4×10^{-2}	1.6, 1.8, 2.5
0.005	widened, $2h$	9.0×10^{-1}	6.2×10^{-1}	3.8×10^{-1}	2.3×10^{-1}	1.1, 1.4, 1.5
0.005	cell mean, h	6.7×10^{-2}	2.0×10^{-2}	5.7×10^{-3}	2.6×10^{-3}	3.6, 3.6, 2.3
0.005	cell mean, $2h$	1.12×10^{-1}	5.3×10^{-2}	2.4×10^{-2}	1.22×10^{-2}	2.2, 2.2, 2.0
0.005	band-limited	8.6×10^{-2}	3.2×10^{-2}	9.9×10^{-3}	3.5×10^{-3}	2.9, 3.3, 3.1
0.01	naive	5.8×10^{-2}	1.78×10^{-2}	5.2×10^{-3}	1.30×10^{-3}	3.5, 3.4, 4.1
0.01	widened, h	3.0×10^{-1}	1.24×10^{-1}	5.2×10^{-3}	1.30×10^{-3}	2.7, 8.9, 4.1
0.01	widened, $2h$	8.4×10^{-1}	5.4×10^{-1}	2.8×10^{-1}	1.15×10^{-1}	1.3, 1.8, 2.6
0.01	cell mean, h	5.4×10^{-2}	1.76×10^{-2}	6.6×10^{-3}	3.1×10^{-3}	3.3, 2.8, 2.2
0.01	cell mean, $2h$	1.09×10^{-1}	5.3×10^{-2}	2.6×10^{-2}	1.31×10^{-2}	2.2, 2.0, 2.0
0.01	band-limited	6.8×10^{-2}	2.2×10^{-2}	6.8×10^{-3}	2.2×10^{-3}	3.3, 3.3, 3.3
floor	naive, uniform	5.3×10^{-2}	1.81×10^{-2}	4.8×10^{-3}	1.30×10^{-3}	3.2, 3.7, 3.9

part inside the band is a smooth superposition of the band's own waves. Our first attempt used the local material's eigenvector, which makes the tractions jump by the impedance ratio, $2.8\times$, over the width δ at every edge crossing; the true dynamics resolve that into $O(1)$ waves with δ -sharp fronts, and both schemes then sat $50\times$ above the resolution floor at first-order rates

TABLE 8. Flat edges at normal incidence: relative ℓ^2 error in σ_{yy} at $t = 1$ and rates, for the runs of Table 3. The floor row is the error in v of the uniform-medium run, repeated for reference.

δ	operator	$n = 2500$	4900	10000	19600	rates
0 (jump)	naive	7.3×10^{-2}	4.1×10^{-2}	2.1×10^{-2}	1.23×10^{-2}	1.7, 1.9, 1.6
0 (jump)	interface-aware	3.9×10^{-2}	1.35×10^{-2}	3.8×10^{-3}	1.07×10^{-3}	3.2, 3.5, 3.8
0.0025	naive	5.9×10^{-2}	4.2×10^{-2}	2.8×10^{-2}	1.23×10^{-2}	1.0, 1.2, 2.4
0.0025	seeds	2.3×10^{-2}	6.8×10^{-3}	2.2×10^{-3}	6.7×10^{-4}	3.6, 3.2, 3.5
0.01	naive	3.9×10^{-2}	1.51×10^{-2}	4.2×10^{-3}	1.00×10^{-3}	2.8, 3.6, 4.3
0.01	seeds	3.7×10^{-2}	8.6×10^{-3}	2.4×10^{-3}	1.07×10^{-3}	4.4, 3.6, 2.4
0.04	naive	3.4×10^{-2}	1.11×10^{-2}	2.9×10^{-3}	7.4×10^{-4}	3.3, 3.8, 4.0
0.04	seeds	4.2×10^{-2}	1.74×10^{-2}	6.0×10^{-3}	2.2×10^{-3}	2.6, 3.0, 3.0
floor	naive, uniform	5.2×10^{-2}	1.83×10^{-2}	4.8×10^{-3}	1.29×10^{-3}	3.1, 3.7, 3.9

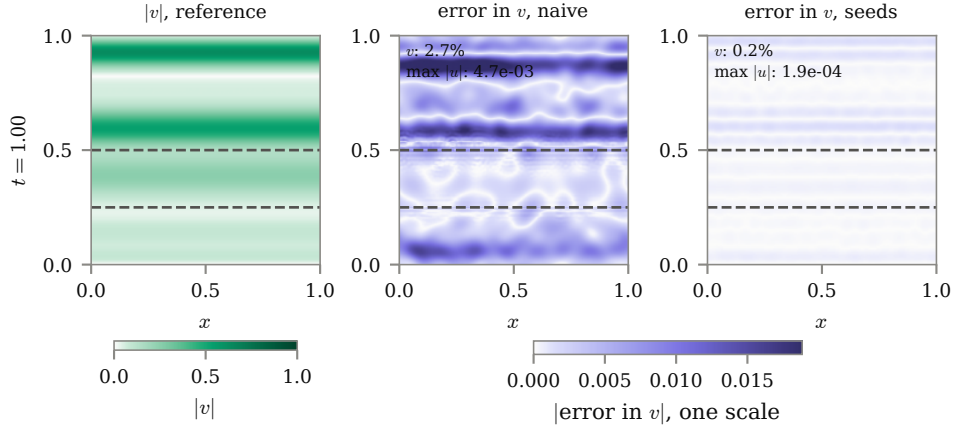


FIGURE 8. The pulse on 10,000 nodes through flat edges of width $\delta = h/4$ at $t = 1$ (the band between the dashed lines): the reference $|v|$, and the error in v of the naive scheme and of the seeds on one colour scale, with each map's relative error and largest spurious $|u|$.

whatever the stencils did at the edge. Initial data must respect the interface conditions at the sub-grid scale, or every scheme measures the initial data.

What is measured. The sweep of Section 6.4 through $\delta = 0.0025$ and 0.01 with the same node sets and the same seed operators, which do not depend on the pulse: relative errors in v , u and σ_{yy} against the Fourier-in- x reference, the two floors, and the ablation of Section 5.3, in which only the seeds of the pure normal monomials y^b are marched and every $x^a y^b$ with $a \geq 1$ is the plain monomial. What the ablation removes is, for instance,

the v -component that the seed of $u = x$ acquires across the edge (0.5 at $\delta = h/4$ on a 400-node stencil): continuity of the normal stress $\lambda u_x + K v_y$ with λ changing forces a kink in v_y , the Poisson coupling of the edge, which a monomial cannot carry.

What the seeds do. Tables 9, 10 and 11 and Figure 9 give the results. The seeds win at every (n, δ) , by less than at normal incidence: $1.4\times$, $1.7\times$, $1.9\times$ and $1.7\times$ in v and $1.2\times$, $1.4\times$, $1.5\times$ and $1.6\times$ in u through $\delta = 0.0025$, and 1.4 to $1.8\times$ in v and 1.1 to $1.7\times$ in u through $\delta = 0.01$, where the naive scheme has $h \leq 2\delta$ and is resolving the edge. They still lead there at $h = 0.71\delta$, by $1.6\times$, so the rule of Section 6.4 is safe on the seeded side here; it was set on v at normal incidence, and this train does not reach below $h = 0.71\delta$. The tangential seeds are what acts, and they act where the edge is sharp. Through $\delta = 0.0025$ the ablation is worse than the full seeds by $1.2\times$, $1.6\times$, $2.2\times$ and $2.3\times$ in v and $1.3\times$, $1.7\times$, $2.4\times$ and $2.7\times$ in u , and it is worse than the naive scheme in u at every n and in v from 10,000 nodes on (1.91×10^{-2} against 1.40×10^{-2} in v at 19,600 nodes, 4.3×10^{-2} against 2.6×10^{-2} in u): the normal-only seeds bend v across the edge for the y^b columns and leave $u = x$ straight, an inconsistent pair that costs more than doing nothing. Through $\delta = 0.01$ the ablation and the full seeds agree to within 4% at every n : with $h \leq 2\delta$ the kink of the tangential columns is spread over half a stencil radius or more, and a cubic in y carries it. So the same crossover $h \approx \delta$ that decides whether to seed decides whether the tangential columns are worth their march.

The order, reported as it is. The order is not four. Through $\delta = 0.0025$ the seeds converge at 2.5, 2.3 and 1.9 in v (the naive scheme at 2.0, 2.0 and 2.2) and sit $7\times$ above the train's resolution floor at 19,600 nodes, where at normal incidence they sat at half the floor; in σ_{yy} they are no better than the naive scheme at 19,600 nodes (1.62×10^{-2} against 1.58×10^{-2} , at rates of 1.3 to 1.5). Six measurements located where this comes from. It is not the hyperviscosity: at $\gamma/4$ the seeds' v moves by under 15% wherever the run stays stable. It is not the angle: the trains along (1, 3), (1, 4) and (1, 6), at 18.4° , 14.0° and 9.5° , give the same rates with a constant that falls with the angle (8.3×10^{-3} , 7.5×10^{-3} , 6.5×10^{-3} and 4.8×10^{-3} at 19,600 nodes from 26.6° to 9.5°). It is not the u -component seeds: an S pulse at normal incidence, which has no x -dependence and no conversion, comes through $\delta = 0.0025$ at rates 4.4, 3.9 and 3.2 in u , at half the naive floor and with a spurious v of 2.7×10^{-5} against the naive scheme's 6.2×10^{-4} at 19,600 nodes, the normal-incidence result in the other polarisation. The jump-aware stencils do the same: the interface-aware operator at $\delta = 0$ with this train, measured against a 78,400-node run of itself resampled one-sided, converges at 2.4, 2.5 and 2.4 in v with 8.9×10^{-3} at 19,600 nodes, and the seeds through $\delta = 0.0025$ (8.3×10^{-3}) are where their $\delta \rightarrow 0$ limit is; they keep the order of the stencils they generalise, an order that was never measured at oblique incidence in [Mar16, MF17a]. The stencils at the edge

TABLE 9. The train at 26.6° through flat edges: relative ℓ^2 error in v at $t = 1$ and rates for the naive scheme, the seeds, the normal-only ablation and the seed floor (the same rows seeded through a contrast of 10^{-6}), with the train’s resolution floor.

δ	operator	$n = 2500$	4900	10000	19600	rates
0.0025	naive	1.17×10^{-1}	6.0×10^{-2}	3.0×10^{-2}	1.40×10^{-2}	2.0, 2.0, 2.2
0.0025	seeds	8.2×10^{-2}	3.6×10^{-2}	1.56×10^{-2}	8.3×10^{-3}	2.5, 2.3, 1.9
0.0025	ablation	9.9×10^{-2}	5.7×10^{-2}	3.5×10^{-2}	1.91×10^{-2}	1.7, 1.4, 1.8
0.0025	seed floor	3.5×10^{-2}	1.26×10^{-2}	3.6×10^{-3}	1.06×10^{-3}	3.1, 3.5, 3.6
0.01	naive	1.01×10^{-1}	4.9×10^{-2}	2.0×10^{-2}	8.2×10^{-3}	2.2, 2.5, 2.7
0.01	seeds	7.4×10^{-2}	3.0×10^{-2}	1.16×10^{-2}	5.1×10^{-3}	2.7, 2.7, 2.4
0.01	ablation	7.3×10^{-2}	3.0×10^{-2}	1.18×10^{-2}	5.3×10^{-3}	2.6, 2.7, 2.4
0.01	seed floor	2.1×10^{-2}	7.2×10^{-3}	3.3×10^{-3}	1.46×10^{-3}	3.1, 2.1, 2.5
floor	naive, uniform	4.7×10^{-2}	1.55×10^{-2}	4.3×10^{-3}	1.20×10^{-3}	3.3, 3.6, 3.8

are not the bottleneck: the truncation error of the elastic operator on the exact reference state at $t = 1$ is fourth order for the bulk rows and for the seed rows at normal incidence, while at oblique incidence the *bulk* rows, nowhere near an edge, sit at 5.8×10^{-2} down to 2.2×10^{-2} , first order, and the seed rows below them. And it is the converted S waves: they keep the P pulse’s temporal spectrum at wavelengths shorter by $c_p/c_s = 1.73$ on both sides of each edge, the spatial profile of an S pulse of sharpness 26, whose own resolution floor in the uniform medium is 2.1×10^{-1} down to 1.19×10^{-2} at rates 2.0, 2.8 and 3.5: pre-asymptotic on every node set here, and at 19,600 nodes above the seeds’ whole error. Weighted by the S waves’ share of the solution it is the seeds’ error, rates included. So at oblique incidence the seeds keep the order of the jump-aware stencils and land at the resolution floor of the mode-converted S waves, which the train’s own floor, a P wave, does not see; the right yardstick for an oblique run is a floor with the converted content in it, and fourth-order behaviour of any scheme on this problem needs h below about $1/300$, the next node set up. We report the result as it is.

The picture. Figure 10 is the train on 10,000 nodes through $\delta = 0.0025 = h/4$ at $t = 1$: the reference’s $|v|$, its curl $|u_y - v_x|$, which is zero for the incident P wave and maps the S waves the edges make, and the two error maps in v on one colour scale. The errors in v and u are 3.0% and 5.5% for the naive scheme and 1.6% and 3.6% for the seeds; at $t = 0.25$ they are 1.2% and 2.0% against 0.9% and 1.5%, both maps showing the edge’s horizontal streaks, the seeds’ fainter, and by $t = 1$ the seeds’ map is about half the naive one everywhere, which is the ratio of the tables.

6.6. The curved edge. The last experiment bends both interfaces into the sine curves of amplitude 0.02 of Section 2.2, curvature up to $\kappa = 0.79$ and tilt

TABLE 10. The train at 26.6° : relative ℓ^2 error in u at $t = 1$ and rates, for the runs of Table 9.

δ	operator	$n = 2500$	4900	10000	19600	rates
0.0025	naive	1.72×10^{-1}	1.06×10^{-1}	5.5×10^{-2}	2.6×10^{-2}	1.4, 1.8, 2.2
0.0025	seeds	1.44×10^{-1}	7.9×10^{-2}	3.6×10^{-2}	1.59×10^{-2}	1.8, 2.2, 2.5
0.0025	ablation	1.86×10^{-1}	1.33×10^{-1}	8.7×10^{-2}	4.3×10^{-2}	1.0, 1.2, 2.1
0.0025	seed floor	3.6×10^{-2}	1.27×10^{-2}	3.6×10^{-3}	1.08×10^{-3}	3.1, 3.5, 3.6
0.01	naive	1.44×10^{-1}	7.6×10^{-2}	3.7×10^{-2}	1.49×10^{-2}	1.9, 2.0, 2.7
0.01	seeds	1.34×10^{-1}	6.3×10^{-2}	2.6×10^{-2}	8.9×10^{-3}	2.2, 2.5, 3.2
0.01	ablation	1.39×10^{-1}	6.4×10^{-2}	2.6×10^{-2}	9.1×10^{-3}	2.3, 2.5, 3.1
0.01	seed floor	2.2×10^{-2}	7.6×10^{-3}	3.4×10^{-3}	1.48×10^{-3}	3.1, 2.2, 2.5
floor	naive, uniform	4.7×10^{-2}	1.55×10^{-2}	4.3×10^{-3}	1.21×10^{-3}	3.3, 3.6, 3.8

TABLE 11. The train at 26.6° : relative ℓ^2 error in σ_{yy} at $t = 1$ and rates, for the runs of Table 9. The seed-floor and floor rows are the errors in v of those runs, repeated for reference.

δ	operator	$n = 2500$	4900	10000	19600	rates
0.0025	naive	1.10×10^{-1}	6.2×10^{-2}	3.1×10^{-2}	1.58×10^{-2}	1.7, 1.9, 2.0
0.0025	seeds	7.0×10^{-2}	4.2×10^{-2}	2.6×10^{-2}	1.62×10^{-2}	1.5, 1.3, 1.4
0.0025	ablation	9.4×10^{-2}	6.3×10^{-2}	4.0×10^{-2}	2.4×10^{-2}	1.2, 1.2, 1.6
0.0025	seed floor	3.5×10^{-2}	1.26×10^{-2}	3.6×10^{-3}	1.06×10^{-3}	3.1, 3.5, 3.6
0.01	naive	9.0×10^{-2}	4.6×10^{-2}	2.2×10^{-2}	9.9×10^{-3}	2.0, 2.1, 2.4
0.01	seeds	7.7×10^{-2}	3.3×10^{-2}	1.31×10^{-2}	6.3×10^{-3}	2.5, 2.6, 2.2
0.01	ablation	8.0×10^{-2}	3.4×10^{-2}	1.33×10^{-2}	6.4×10^{-3}	2.6, 2.6, 2.2
0.01	seed floor	2.1×10^{-2}	7.1×10^{-3}	3.3×10^{-3}	1.46×10^{-3}	3.2, 2.1, 2.4
floor	naive, uniform	4.7×10^{-2}	1.55×10^{-2}	4.3×10^{-3}	1.20×10^{-3}	3.3, 3.6, 3.8

up to 7.2° , with edges of width δ in the true signed normal distance and the seeds marched along the true normal, route (a) of Section 5.4. Everything is at normal incidence, which is locally oblique by up to 7.2° , so the tangential seeds of Section 6.5 act on every stencil not at a crest or a trough, while mode conversion at 7° is weak and the converted-S floor of that section should be small. The node sets are those of Section 2.2 with their rows straddling the sine curves, independent of δ ; the widths are $\delta = 0, 0.005, 0.01$ and then 0.0025; the seeded rows number 1320, 2452, 4631 and 8553 at $\delta = 0.005$ (53% down to 44% of the nodes) and match the flat counts to within a percent at 0.0025 and 0.01; the references are the product-grid solver and the 122,500-node jump-aware run of Section 6.1. Two probes are added. The error in u is now the relative error against the reference's own u , which the curved edges make out of the incident pulse (0.12 at most), so it measures the scattered field alone. And the *truncation probe* applies each operator to the reference state at $t = 1$ on the nodes and compares with

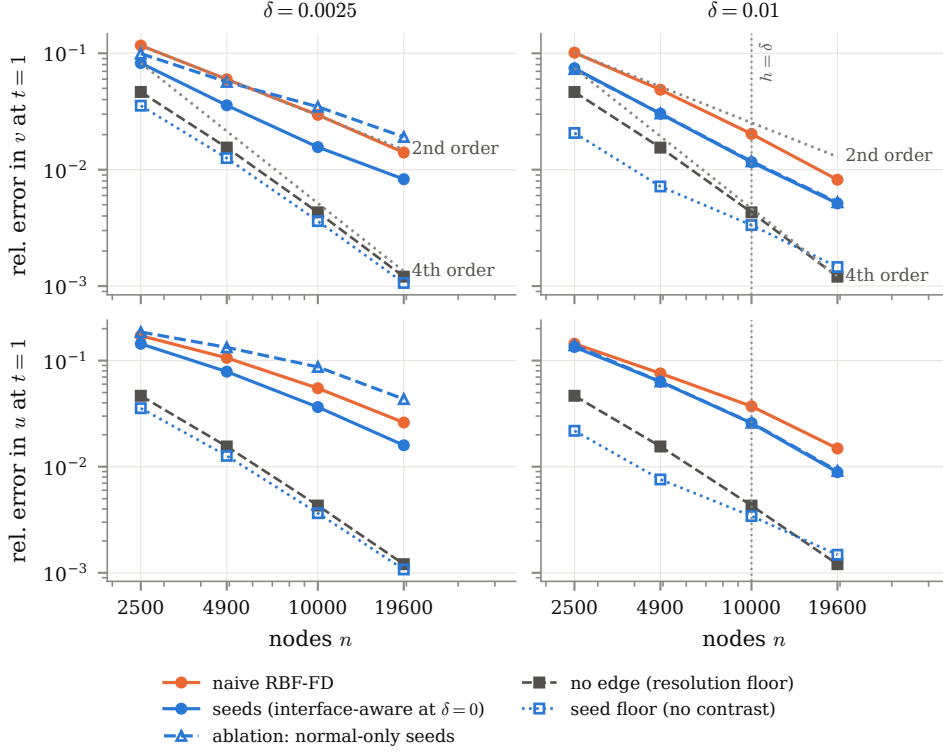


FIGURE 9. The oblique sweep of Tables 9 and 10: relative error in v (upper panels) and in u (lower panels) against n through $\delta = 0.0025$ and 0.01 , for the naive scheme, the seeds and the normal-only ablation, with the train’s resolution floor and the seed floor. Every scheme converges at about second order through the sharp edge, and the ablation crosses above the naive scheme.

the exact rate from the reference’s own derivatives and the material at the nodes, as a relative ℓ^2 error over the seeded rows and over the rest, for both operators (the naive operator’s “edge rows” are the rows the seeds would rebuild); this is the direct measurement of what route (a)’s geometry costs the stencils, with no time stepping in between.

The jump stencils, and the edge the nodes do not resolve. Tables 12, 13 and 14 and Figure 11 give the sweep. The curved jump stencils of [MF17a] are at the resolution floor at every n , 0.93 to $1.07\times$ it, at rates 3.3 , 3.6 , 3.6 , with the naive scheme $14\times$ worse at $19,600$ nodes. That is the measurement Section 6.5 asked for before the curved seeds could be judged: at tilts up to 7.2° the locally flat jump construction keeps the order it has on a flat interface through $19,600$ nodes, against a reference of its own kind with a 5% bar, and the converted-S floor that took a third of the order away at 26.6°

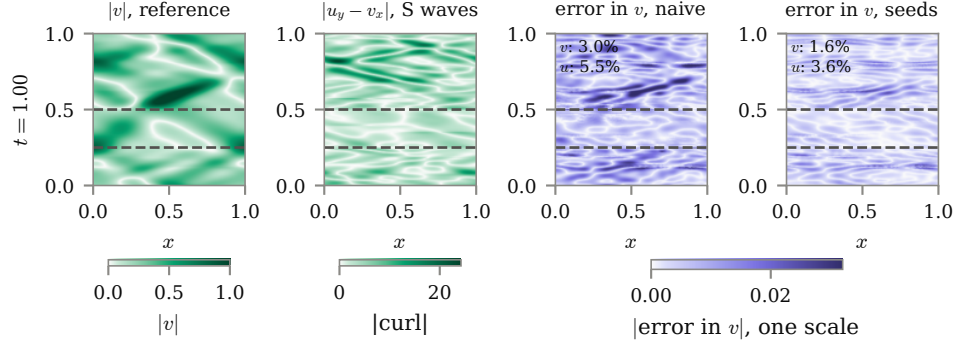


FIGURE 10. The train at 26.6° on 10,000 nodes through flat edges of width $\delta = h/4$ at $t = 1$: the reference $|v|$, its curl $|u_y - v_x|$ (the mode-converted S waves), and the error in v of the naive scheme and of the seeds on one colour scale, with each map's relative errors in v and u .

does not register at 7° . Through $\delta = 0.005$, h/δ from 4 to 1.4, the curved seeds are at their own floor at every n ($0.79\times$, $0.96\times$, $1.2\times$ and $1.06\times$ the seed floor) and $3.1\times$, $7.0\times$, $3.1\times$ and $2.2\times$ below the naive scheme, which has its knee at $h \approx 2\delta$ (rates 2.8, 4.5, 3.5: it resolves the edge from 10,000 nodes on and is still above the seeds at $h = 1.4\delta$); in σ_{yy} the seeds are $3.8\times$, $7.7\times$, $5.1\times$ and $2.9\times$ below it. Through $\delta = 0.01$ the seeds' curved numbers are the flat ones to two digits from 4900 nodes on (1.04×10^{-2} , 3.9×10^{-3} , 1.62×10^{-3} against 1.05×10^{-2} , 3.9×10^{-3} , 1.57×10^{-3}) and the naive scheme's are within 17% of them, and the rule reads off the same column as before: at $h = \delta$ the seeds win by $1.35\times$, at $h = 0.71\delta$ the naive scheme wins by $1.25\times$, the flat crossover. Curvature at amplitude 0.02 changed nothing that these node sets can see.

The seeds' floor. The seed floor is below the naive floor at 4900 and 10,000 nodes (4.7×10^{-3} against 1.81×10^{-2} and 1.68×10^{-3} against 4.8×10^{-3} at $\delta = 0.005$) and above it at 19,600 (8.3×10^{-4} against 1.30×10^{-3}), with its rates falling to 2.1 to 2.6 there: 19-node degree-three rows are third order in the first derivatives with a small constant, and with exact inequality they cover the 19δ tails on both sides of both edges, 44 to 72% of the nodes. That is the mechanism behind the seeds' rates of 2.3 to 2.6 at the fine end, here and in Sections 6.4 and 6.5: a resolution effect of the seeded region's stencils and not of the edge, which the trim below confirms. The scattered field says the same from the other side. In u the jump stencils are $4.2\times$ better than the naive scheme at 19,600 nodes (6.6×10^{-3} against 2.8×10^{-2}) at rates 2.5, 3.1, 3.4 against 2.0, 2.2, 1.6; through $\delta = 0.005$ the seeds are $2.3\times$ better (2.6×10^{-3} against 6.0×10^{-3}) at 3.4, 3.5, 3.7; through $\delta = 0.01$ the two agree at 19,600 nodes (3.9×10^{-3} against 4.4×10^{-3}), where the naive scheme resolves the edge. The seeds' u converges faster than their v

TABLE 12. Curved edges (amplitude 0.02) at normal incidence: relative ℓ^2 error in v at $t = 1$ and rates, for the naive scheme, the seeds marched along the true normal (the interface-aware stencils at $\delta = 0$) and the seed floor, with the resolution floor. h/δ runs from 8 to 2.9 at $\delta = 0.0025$, from 4 to 1.4 at 0.005 and from 2 to 0.71 at 0.01.

δ	operator	$n = 2500$	4900	10000	19600	rates
0 (jump)	naive	1.25×10^{-1}	6.8×10^{-2}	3.2×10^{-2}	2.0×10^{-2}	1.8, 2.1, 1.4
0 (jump)	interface-aware	5.1×10^{-2}	1.69×10^{-2}	4.6×10^{-3}	1.39×10^{-3}	3.3, 3.6, 3.6
0.0025	naive	9.2×10^{-2}	4.6×10^{-2}	2.4×10^{-2}	1.19×10^{-2}	2.1, 1.9, 2.1
0.0025	seeds	2.9×10^{-2}	8.0×10^{-3}	2.9×10^{-3}	9.5×10^{-4}	3.8, 2.9, 3.3
0.0025	seed floor	2.7×10^{-2}	6.7×10^{-3}	1.93×10^{-3}	7.1×10^{-4}	4.2, 3.5, 3.0
0.005	naive	8.1×10^{-2}	3.2×10^{-2}	6.3×10^{-3}	1.90×10^{-3}	2.8, 4.5, 3.5
0.005	seeds	2.6×10^{-2}	4.5×10^{-3}	2.0×10^{-3}	8.8×10^{-4}	5.2, 2.3, 2.5
0.005	seed floor	3.3×10^{-2}	4.7×10^{-3}	1.68×10^{-3}	8.3×10^{-4}	5.8, 2.9, 2.1
0.01	naive	5.8×10^{-2}	1.78×10^{-2}	5.2×10^{-3}	1.30×10^{-3}	3.5, 3.4, 4.1
0.01	seeds	4.2×10^{-2}	1.04×10^{-2}	3.9×10^{-3}	1.62×10^{-3}	4.2, 2.8, 2.6
0.01	seed floor	5.1×10^{-2}	1.20×10^{-2}	3.8×10^{-3}	1.57×10^{-3}	4.3, 3.2, 2.6
floor	naive, uniform	5.3×10^{-2}	1.81×10^{-2}	4.8×10^{-3}	1.30×10^{-3}	3.2, 3.7, 3.9

TABLE 13. Curved edges: relative ℓ^2 error in u at $t = 1$ and rates, for the runs of Table 12. The curved edges make u out of the incident pulse, so its relative error measures the scattered field; the floors' pulse has no u , and their rows are the largest spurious $|u|$.

δ	operator	$n = 2500$	4900	10000	19600	rates
0 (jump)	naive	2.0×10^{-1}	1.04×10^{-1}	4.8×10^{-2}	2.8×10^{-2}	2.0, 2.2, 1.6
0 (jump)	interface-aware	1.46×10^{-1}	6.4×10^{-2}	2.1×10^{-2}	6.6×10^{-3}	2.5, 3.1, 3.4
0.0025	naive	1.89×10^{-1}	1.04×10^{-1}	4.4×10^{-2}	2.0×10^{-2}	1.8, 2.4, 2.4
0.0025	seeds	1.16×10^{-1}	4.3×10^{-2}	1.40×10^{-2}	4.0×10^{-3}	2.9, 3.2, 3.7
0.0025	seed floor	1.85×10^{-3}	7.2×10^{-4}	2.4×10^{-4}	6.7×10^{-5}	2.8, 3.0, 3.8
0.005	naive	1.77×10^{-1}	7.8×10^{-2}	2.0×10^{-2}	6.0×10^{-3}	2.4, 3.9, 3.5
0.005	seeds	1.02×10^{-1}	3.3×10^{-2}	9.3×10^{-3}	2.6×10^{-3}	3.4, 3.5, 3.7
0.005	seed floor	2.2×10^{-3}	1.15×10^{-3}	3.5×10^{-4}	8.0×10^{-5}	1.9, 3.4, 4.3
0.01	naive	1.24×10^{-1}	4.9×10^{-2}	1.53×10^{-2}	4.4×10^{-3}	2.8, 3.3, 3.7
0.01	seeds	9.9×10^{-2}	3.2×10^{-2}	9.7×10^{-3}	3.9×10^{-3}	3.3, 3.3, 2.7
0.01	seed floor	5.0×10^{-3}	1.88×10^{-3}	5.7×10^{-4}	1.37×10^{-4}	2.9, 3.4, 4.2
floor	naive, uniform	9.5×10^{-4}	3.9×10^{-4}	1.20×10^{-4}	2.6×10^{-5}	2.7, 3.3, 4.6

at $\delta = 0.005$ (3.4 to 3.7 against 2.3 to 2.5) because the v floor at the fine end is the seeded rows' resolution of the *incident* pulse, which u , born at the edge, does not carry.

What route (a) costs. Table 15 is the truncation probe. On the true curved solution the seed rows' error is 2.0 to $3.6 \times$ the bulk rows' at $\delta = 0.005$ and

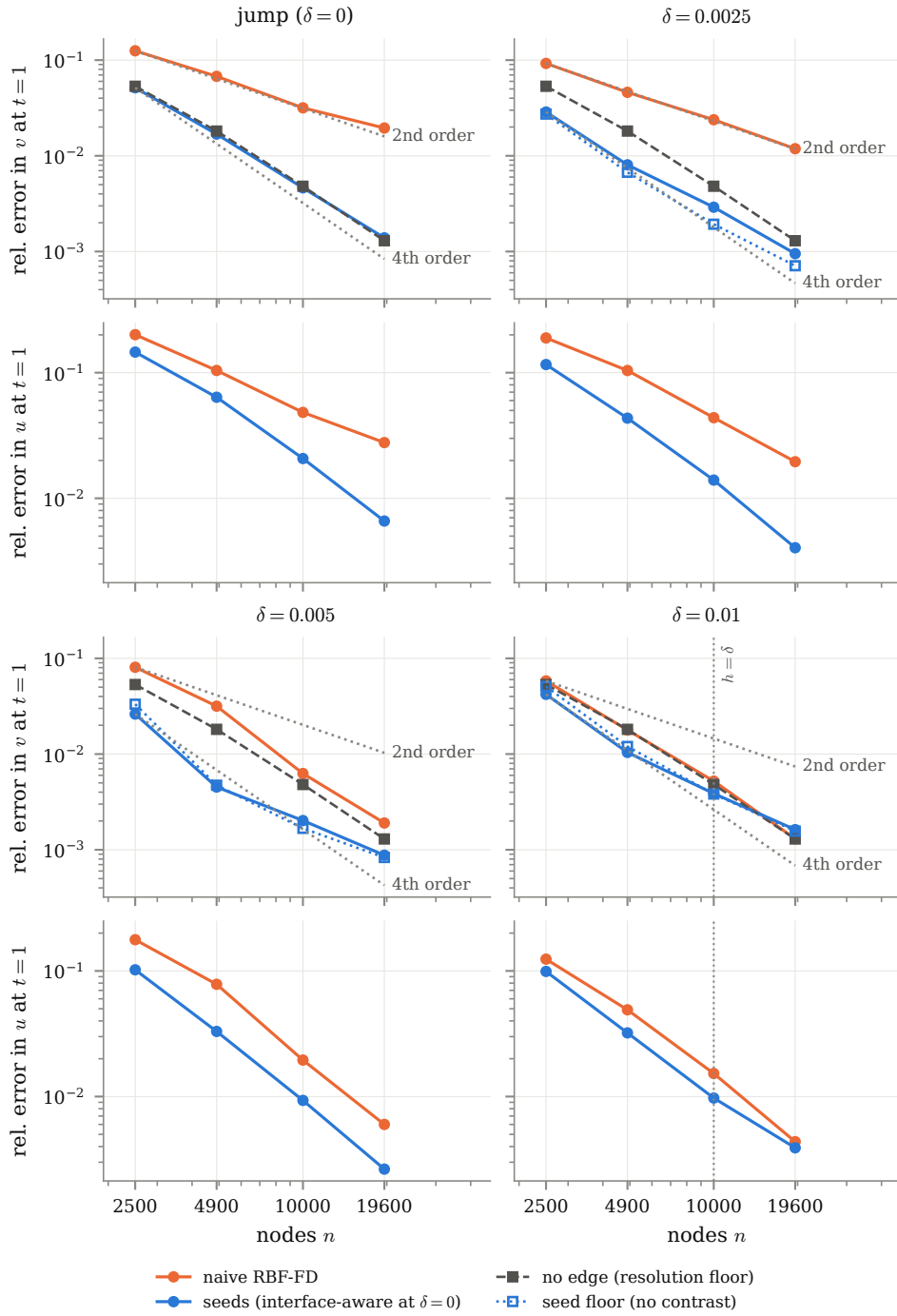


FIGURE 11. The curved sweep of Tables 12 and 13: relative error in v (upper panels) and in u (lower panels) against n for the jump and three edge widths, naive scheme against seeds, with the resolution floor and the seed floor; the vertical line in the $\delta = 0.01$ panel is the node set at which $h = \delta$. The jump stencils lie on the floor; through $\delta = 0.0025$ and 0.005 the seeds lie on the seed floor.

TABLE 14. Curved edges: relative ℓ^2 error in σ_{yy} at $t = 1$ and rates, for the runs of Table 12. The seed-floor and floor rows are the errors in v of those runs, repeated for reference.

δ	operator	$n = 2500$	4900	10000	19600	rates
0 (jump)	naive	9.8×10^{-2}	4.7×10^{-2}	2.0×10^{-2}	1.28×10^{-2}	2.2, 2.4, 1.4
0 (jump)	interface-aware	4.6×10^{-2}	1.62×10^{-2}	4.4×10^{-3}	1.28×10^{-3}	3.1, 3.6, 3.7
0.0025	naive	8.5×10^{-2}	4.5×10^{-2}	2.5×10^{-2}	1.16×10^{-2}	1.9, 1.7, 2.3
0.0025	seeds	2.6×10^{-2}	8.8×10^{-3}	2.8×10^{-3}	8.0×10^{-4}	3.3, 3.2, 3.7
0.0025	seed floor	2.7×10^{-2}	6.7×10^{-3}	1.93×10^{-3}	7.1×10^{-4}	4.2, 3.5, 3.0
0.005	naive	8.4×10^{-2}	3.4×10^{-2}	6.9×10^{-3}	1.79×10^{-3}	2.7, 4.4, 4.0
0.005	seeds	2.2×10^{-2}	4.4×10^{-3}	1.35×10^{-3}	6.1×10^{-4}	4.8, 3.3, 2.4
0.005	seed floor	3.3×10^{-2}	4.7×10^{-3}	1.66×10^{-3}	8.3×10^{-4}	5.8, 2.9, 2.1
0.01	naive	5.4×10^{-2}	1.74×10^{-2}	4.5×10^{-3}	1.14×10^{-3}	3.4, 3.8, 4.1
0.01	seeds	3.9×10^{-2}	8.5×10^{-3}	2.6×10^{-3}	1.11×10^{-3}	4.6, 3.3, 2.5
0.01	seed floor	5.1×10^{-2}	1.20×10^{-2}	3.8×10^{-3}	1.57×10^{-3}	4.3, 3.2, 2.6
floor	naive, uniform	5.3×10^{-2}	1.81×10^{-2}	4.8×10^{-3}	1.30×10^{-3}	3.2, 3.7, 3.9

0.01 and converges at the bulk rows' rate (3.2 to 3.9 against 3.1 to 3.7), with no flattening at the fine end, where $\kappa r^2/2$ is 2.6% of $\delta = 0.005$; the naive edge rows are 4 to $24\times$ the seeds' at $\delta = 0.005$ and converge at 1.5 to 2.4 until the edge is resolved. A degree-three 19-node row is expected to sit a factor above a degree-four 30-node row on a smooth solution, and the factor is the same at both widths, so nothing of it is the curvature. Through $\delta = 0.0025$, the case the estimate of Section 5.4 singled out ($\kappa r^2/2$ is 40%, 20%, 10% and 5% of δ on the four node sets), three things happen. The headline holds: through an edge the nodes never resolve, on curved interfaces, the seeds are $3.2\times$, $5.7\times$, $8.2\times$ and $12.5\times$ below the naive scheme in v ($14\times$ in σ_{yy} and $4.8\times$ in u at 19,600 nodes), at rates 3.8, 2.9 and 3.3 against 2.1, 1.9 and 2.1, and at 19,600 nodes their 9.5×10^{-4} is below the naive scheme's floor with no edge at all (1.30×10^{-3}). This is also where route (a)'s geometry first shows: the curved seeds are $0.95\times$, $1.04\times$, $1.4\times$ and $1.5\times$ the flat seeds on the four node sets, they sit at $1.05\times$, $1.2\times$, $1.5\times$ and $1.3\times$ their own floor where through $\delta = 0.005$ they sat at 0.8 to $1.2\times$, and the seed rows' truncation ratio to the bulk rows *rises* with n , from 4.3 to 4.8, where through $\delta = 0.005$ it fell from 3.6 to 2.8: a term that shrinks as h^2 against one that shrinks as $h^{3.5}$. Taking the excess over the seed floor in quadrature, the geometric contribution at 19,600 nodes is 6.3×10^{-4} , half the naive floor and level with the seed floor; it would lead at the next node set. And the estimate was the right size: 5% of δ in the material at the far nodes of a stencil costs the seeds half their error at 19,600 nodes, and 40% of δ at 2500 nodes costs nothing, because the resolution error there is $30\times$ larger.

The trim. Table 16 seeds only the rows whose 19-node stencil sees more than 10^{-3} of the contrast, the tanh tails to 3.8δ instead of 19δ : 601, 977,

TABLE 15. The truncation probe on the curved geometry: relative ℓ^2 error of the elastic operator applied to the reference state at $t = 1$, over the rows the seeds rebuild (for the naive operator and for the seeds) and over the remaining rows (the same 30-node degree-four stencils in both operators), and the ratio of the seed rows' error to the bulk rows'.

δ	n	naive, edge rows	seeds, edge rows	bulk rows	seeds/bulk
0.0025	2500	5.9×10^{-2}	1.08×10^{-2}	2.5×10^{-3}	4.3
0.0025	4900	4.8×10^{-2}	3.6×10^{-3}	8.3×10^{-4}	4.4
0.0025	10000	3.6×10^{-2}	1.07×10^{-3}	2.4×10^{-4}	4.5
0.0025	19600	2.2×10^{-2}	3.3×10^{-4}	7.0×10^{-5}	4.8
0.005	2500	3.9×10^{-2}	9.0×10^{-3}	2.5×10^{-3}	3.6
0.005	4900	2.3×10^{-2}	2.5×10^{-3}	8.6×10^{-4}	2.9
0.005	10000	1.05×10^{-2}	6.3×10^{-4}	2.5×10^{-4}	2.5
0.005	19600	4.7×10^{-3}	2.0×10^{-4}	7.1×10^{-5}	2.8
0.01	2500	1.83×10^{-2}	6.1×10^{-3}	2.6×10^{-3}	2.4
0.01	4900	8.8×10^{-3}	1.79×10^{-3}	9.0×10^{-4}	2.0
0.01	10000	3.1×10^{-3}	5.0×10^{-4}	2.5×10^{-4}	2.0
0.01	19600	7.5×10^{-4}	1.66×10^{-4}	7.3×10^{-5}	2.3

1585 and 2746 rows at $\delta = 0.005$ (24% down to 14% of the nodes) in place of 1320 to 8553. The trim does for the rate what the floor paragraph predicted: 3.4, 3.2, 3.6 all the way, no decay at the fine end, and the trimmed rows keep rates of 3.4 to 4.0 in the truncation probe. It does not lower the error on these node sets: at 19,600 nodes the two are equal in v (9.4×10^{-4} against 8.8×10^{-4} , both below the naive floor), and at every coarser n the untrimmed seeds are 1.2 to $2.2\times$ better in v and up to $1.5\times$ better in u , because the 19-node degree-three rows have the smaller error constant there and the wide seeded region gets that constant on half the domain. So the fine-end rate of the untrimmed seeds is confirmed to be the seeded region's own resolution and not the edge, the trim is the lever that restores the order beyond 19,600 nodes (extrapolating the two rates, the trimmed seeds lead by 40% at 40,000 nodes), and where between 3.8δ and 19δ to put the seeded region's boundary is a tuning question for larger node sets (Section 7).

Stability, curved. Table 17 and Figure 12 repeat the acceptance run of Section 6.3 on the curved geometry, the seeds on their true normals. Row for row it is the flat table: $+6.0 \times 10^{-2}$, $+8.3 \times 10^{-2}$ and $+1.05 \times 10^{-1}$ against $+6.3 \times 10^{-2}$, $+7.8 \times 10^{-2}$ and $+9.6 \times 10^{-2}$, at max $|R|$ of 1.0003 to 1.0004 and energy ratios 0.988, 0.999 and 1.016; the jump-aware operator at $+4.7 \times 10^{-2}$ and 1.0002. The rotation of every seed stencil into its own frame, the tilted normals and the curved rows change the rightmost eigenvalues by at most 10% and the RK4 amplification not at all: the answer of Section 6.3 carries over to the curved edge unchanged, with the 30-node Δ^3 footprint on the seed rows.

TABLE 16. The trimmed seeds through curved edges of width $\delta = 0.005$: relative ℓ^2 error in v at $t = 1$ and rates with the seeded rows trimmed to the stencils that see more than 10^{-3} of the contrast, against the naive scheme and the resolution floor. The untrimmed seeds are the $\delta = 0.005$ row of Table 12.

δ	operator	$n = 2500$	4900	10000	19600	rates
0.005	naive	8.1×10^{-2}	3.2×10^{-2}	6.3×10^{-3}	1.90×10^{-3}	2.8, 4.5, 3.5
0.005	seeds	3.2×10^{-2}	1.00×10^{-2}	3.2×10^{-3}	9.4×10^{-4}	3.4, 3.2, 3.6
floor	naive, uniform	5.3×10^{-2}	1.81×10^{-2}	4.8×10^{-3}	1.30×10^{-3}	3.2, 3.7, 3.9

TABLE 17. The acceptance run of Table 6 on the curved geometry (2500 nodes, amplitude 0.02): rebuilt rows, the spectrum's largest real part with the hyperviscosity in, the largest RK4 amplification and the energy ratio of the run to $t = 1$.

δ	operator	rebuilt rows	max Re λ , with γ	RK4 max $ R $	$E(1)/E(0)$
jump	naive	0	$+4.7 \times 10^{-3}$	1.00002	0.955
jump	interface-aware	793	$+4.7 \times 10^{-2}$	1.00019	0.970
$h/8$	naive	0	$+4.7 \times 10^{-3}$	1.00002	0.956
$h/8$	seeds	982	$+6.0 \times 10^{-2}$	1.00025	0.988
$h/2$	naive	0	$+3.8 \times 10^{-3}$	1.00002	0.982
$h/2$	seeds	1796	$+8.3 \times 10^{-2}$	1.00034	0.999
$2h$	naive	0	$+2.1 \times 10^{-5}$	1.00000	0.991
$2h$	seeds	2500	$+1.05 \times 10^{-1}$	1.00043	1.016

The picture. Figure 13 is the run on 10,000 nodes through curved edges of width $\delta = 0.005 = h/2$ at $t = 1$: the reference's $|v|$ from the product grid, its curl, which maps the S waves the curved edges make out of the incident P pulse, and the two error maps in v on one colour scale. The errors in v and u are 0.63% and 2.0% for the naive scheme and 0.20% and 0.93% for the seeds. At $t = 0.25$ they are 0.37% and 1.6% against 0.04% and 0.5%, and the naive map is then the picture of the problem: a streak that follows the sine of the upper edge, darkest where the pulse has just arrived at the trough, with the transmitted pulse's error below it, while the seeds' map has no streak, only the resolution speckle above the band, ten times lower in norm. By $t = 1$ both maps are the propagated pulse's own resolution error, and the seeds' is a third of the naive one, which is the tables' $3.1 \times$.

The route (b) decision, and cost. On this evidence route (b) of Section 5.4, the boundary-value problem per stencil, is not needed on these node sets: through $\delta = 0.005$ and 0.01 the route (a) seeds sit at the seed operator's own floor at every n , their rows' truncation error on the true curved solution converges at the bulk rate, through $\delta = 0.0025$ the geometry first shows as

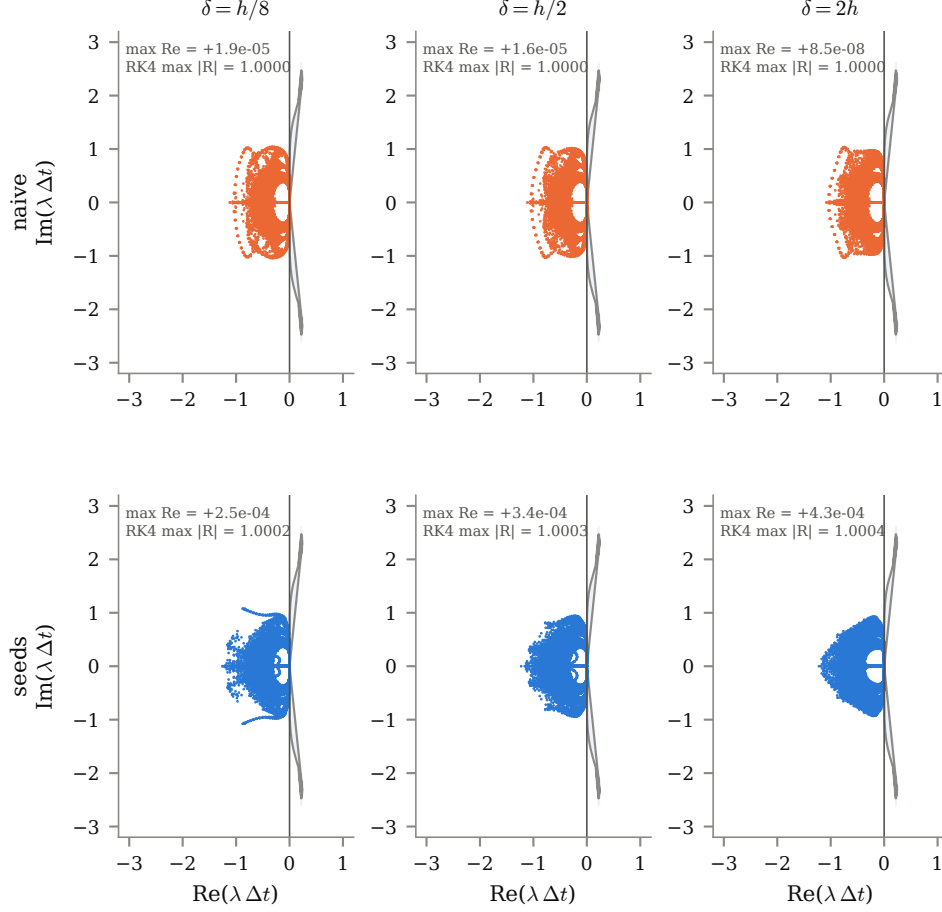


FIGURE 12. Spectra of the naive (top) and seed (bottom) operators on the 2500-node curved set through edges of width $\delta = h/8$, $h/2$ and $2h$, as in Figure 6.

1.4 to $1.5\times$ the flat seeds' error at 10,000 and 19,600 nodes, a contribution level with the seed floor and half the naive floor, and the curved sweep reproduces the flat one. The floor route (a) does impose is the geometric one, $\kappa r^2/2$ against δ , and it is below the resolution floor of these node sets at every width run. What would make (b) necessary is a radius of curvature comparable to the stencil, or a δ so small that $\kappa r^2/2\delta$ is of order one at a resolution the node set can still afford; neither is this geometry, and if either arises the cheaper first step is the curvilinear evaluation of the same marched seeds (Sections 5.4 and 7). Two things were not done: the oblique train on the curved geometry, which the product-grid reference could take as it takes any initial state, and the tuning of the seeded width. The costs, for the record: the flat sweep takes 636s on 12 workers with the seed operators

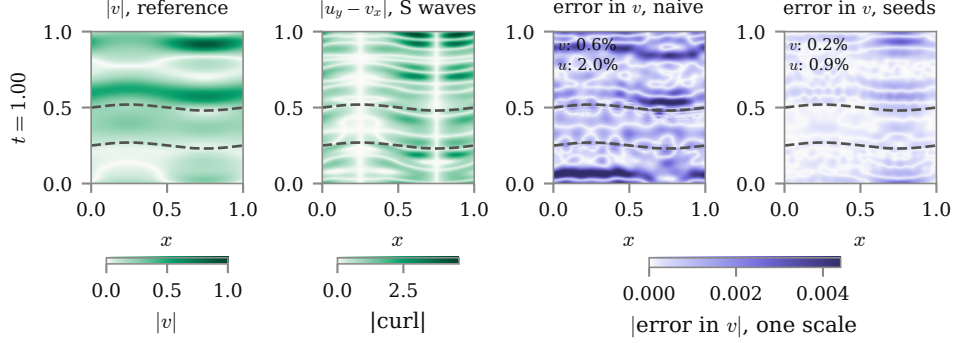


FIGURE 13. The pulse on 10,000 nodes through curved edges of width $\delta = h/2$ at $t = 1$ (the interfaces dashed): the reference $|v|$, its curl $|u_y - v_x|$ (the S waves made at the curved edges), and the error in v of the naive scheme and of the seeds on one colour scale, with each map's relative errors in v and u .

to build (the largest, 19,600 nodes at $\delta = 0.04$, in 94 s) and about 4 minutes once they are cached; the oblique sweep 500 s with the operators cached, 260 s of it the two references; on the curved geometry the references took 45 minutes plus 3.6 hours for the 2048×4096 grid, the 24 seed operators about an hour on 6 to 12 workers (the two 19,600-node operators 330 and 475 s on six), the sweep 12 minutes from the caches and the spectra 23 minutes. A curved march costs $3.6\times$ a flat one (Section 5.4), and none of this is tuned.

7. DISCUSSION

7.1. Limitations. Section 2.4 lists what the implementation leaves out of the 2016–2017 methods. The restrictions below are those of the experiments.

Order and contrast. Everything was run at fourth order, FD4 in one dimension and degree four away from and degree three across an edge in two, at one material contrast in each dimension (Section 2) and one pulse per test problem. Nothing was tried at a sharper contrast, for a layer thinner than the 19δ tails of an edge, or at an order above four (Section 4). The construction does not change for any of them, and the conditioning of Section 3.1, the stability at the standard γ of Section 6.3 and the seeded width were measured only where the tables say they were. In one dimension an edge below about $10^{-5}h$ is better served by the interface-aware weights directly, since the seeds converge to them at first order in δ (Section 3.5).

Node sets. The largest node set has 19,600 nodes, and the fine end of every two-dimensional sweep is a resolution effect of the seeded region, not of the edge: with no tolerance on the material spread the seeded rows cover

the 19δ tails on both sides of both edges, 44 to 72% of the nodes, with 19-node degree-three rows whose first derivatives are third order, so the seeds' rates fall to 2.3 to 2.6 at the fine end and the seed floor's own rate to 2.0 between 10,000 and 19,600 nodes at $\delta = 0.005$ (Section 6.6). The trim to 3.8δ restores rates of 3.2 to 3.6 without lowering the error on these node sets. Where between 3.8δ and 19δ to put the boundary of the seeded region beyond 19,600 nodes was not settled, and no result here says what the seeds do on a node set on which the trim would matter.

The rule's evidence. The rule of Section 6.4 was set on v at normal incidence through flat edges and read off the same column on the curved geometry (Section 6.6). At oblique incidence it was checked on the seeded side only: the train does not reach below $h = 0.71\delta$, where the seeds still lead by $1.6\times$ (Section 6.5), so nothing at oblique incidence says where the seeds stop paying. The mode-converted S waves put every scheme, the jump stencils included, at order about 2.5 at 26.6° on these node sets and do not register at 7° ; fourth-order behaviour at oblique incidence on this problem needs h below about $1/300$, the next node set up. The energy growth of the seed operator, about 0.012 per unit time at $\delta = 2h$ (Section 6.3), was found only in the regime the rule excludes, in which every row is a seed row, and was not chased.

Geometry. Route (a) evaluates seeds marched along one normal in the tangent frame of the stencil, so the material at a stencil's far nodes is wrong by $\kappa r^2/2$ against δ (Section 5.4). On the two sine interfaces run, at amplitude 0.02, that error first shows through $\delta = 0.0025$ at 10,000 and 19,600 nodes, as 1.4 to $1.5\times$ the flat seeds' error, level with the seed floor, and would lead at the next node set (Section 6.6). A radius of curvature comparable to the stencil was not run, nor was the oblique train on the curved geometry.

Cost. Each seed row is one or two marches of the 600-state system of Section 5.2: 21 to 25 ms for a 19-node stencil and 53 ms for a seed row on a flat edge, $3.6\times$ that on a curved one, and the two largest curved operators took 330 and 475 s on six workers (Sections 5.3 and 6.6). Neither the flat translation symmetry, one march per row with dense output, nor the 19-node stencil as the prefix of the 30-node one was used, and nothing is tuned; the operators are cached, and the references of the curved geometry cost more than everything else together (45 minutes plus 3.6 hours). These are the costs of the reference implementation and an upper bound on the construction's.

The comparison. The coefficient treatments of Sections 4 and 6 are our implementations, one scheme per dimension, one contrast and one pulse, and their sources analyse and run them on staggered Cartesian grids. The one-dimensional grid is cell-centred with every jump on a cell boundary, which is why the one-cell mean changes nothing at a jump there; the two-dimensional averaging is an isotropic reading of the compliance rule, the wave moduli harmonically and the density arithmetically over a square; and

the anisotropic effective medium of Schoenberg and Muir, which [KWR22] find best in elastic media, was not built because the implementation has no anisotropic five-field operator. The measurement licenses what Section 1.1 states and no more; it is not a ranking of the methods in general.

7.2. Future work. In the order we would do them.

- (1) *Curvilinear evaluation of the marched seeds*, before route (b) of Section 5.4: keep the marches, evaluate each seed at the node's true arclength and normal distance instead of its tangent coordinates, and scale the tangential jets at the anchor by $1/(1 - \kappa y_e)$. The material is then exact at every node at no march cost and only the operator's curvature terms remain. It is worth building where route (a)'s geometric floor, $\kappa r^2/2$ against δ , meets the resolution floor: the next node set up through $\delta = 0.0025$, or a tighter curve.
- (2) *The seeded width*. The trim of Section 6.6 is the lever that restores the order beyond 19,600 nodes (extrapolating the two rates, the trimmed seeds lead by 40% at 40,000 nodes); where between 3.8δ and 19δ to put the boundary of the seeded region is one sweep at the next node set.
- (3) *The seeds' true Δ^3 as hyperviscosity right-hand sides*. The hyperviscosity rows of Section 5.3 impose $\Delta^3 S = 0$ on every seed, a choice; the alternative is the seeds' actual sixth derivatives at the anchor, which need the normal derivatives of the coefficient functions to sixth order and so the material's to fifth. Finite differences of the marched seeds cannot give them at a sharp edge (a sixth difference at step $\delta/10$ amplifies rounding by 10^{13}); a Taylor recursion on the first-order system at the anchor, with the Taylor coefficients of the coefficient matrix from the tanh recurrence $T' = (1 - T^2)/\delta$, can, in about sixty lines plus a test against the monomials' Δ^3 in constant material. What it would do is predictable: at an anchor a distance d from an edge the sixth derivative scales like $(r_{\max}/\delta)^5 e^{-2d/\delta}$, so the hyperviscosity would act on the seed part of the solution with strength $\gamma \Delta^3 S \propto (h/\delta)^5$, a consistent discretisation of $\gamma \Delta^3$ applied to the edge structure, strong exactly when the edge is unresolved. That is the behaviour the seeds exist to remove, so we expect a stable operator with more damping, not less, that damps the resolved physics at the edge, and would build it only if the spectrum or the errors ever called for it.
- (4) *The cost of the marches*. The two optimisations Section 5.3 names, one march per row on a flat edge and one march per stencil for both the 19- and the 30-node rows; and in one dimension Mühlbach's recurrence for generalised divided differences [M73], the analogue of Fornberg's algorithm for an extended complete Chebyshev system, in place of the dense 5×5 solve. None was needed for the sweeps.

- (5) *Two runs the drivers can take today.* The oblique train on the curved geometry, since the product-grid reference of Section 6.1 takes any initial state, to see whether the converted-S floor of Section 6.5 moves with curvature; and a dispersion study in a uniform medium of the 30-node annihilating Δ^3 rows, to settle why the band of seed rows is more accurate on this pulse than 30-node degree-four stencils everywhere (Section 6.4), on which nothing here depends.
- (6) *What the construction does not change for.* Higher order, which in two dimensions meets the instability Section 5.3 found for degree-four rows across a sharp edge; other contrasts; an edge or a layer thinner than the 19δ tails, where a stencil sees both edges of a band and the implementation has no triple stencils (Section 2.4); the variable Lamé parameters of the published test case, which vary along the edge as well as across it, so that the ansatz of Section 5.1, the material a function of the normal coordinate alone, no longer applies as it stands; and three dimensions, where the ansatz has two tangential coordinates and the same normal one. The implementation is two-dimensional and none of these was run.
- (7) *From the comparison.* An anisotropic five-field operator, which would let the Schoenberg–Muir medium that [KWR22] find best in elastic media run head to head with the seeds on the same nodes; and the seeds on a smoothed medium: applied to the medium the coefficient treatments manufacture, a seeded scheme would be expected to remove the second-order component those treatments accept at the interface, which is a hypothesis and nothing here tested.

7.3. Conclusions. Between a medium that varies on the scale of the grid and one with a jump lies the edge that is smooth but thinner than the grid, and the question was whether a stencil on an unchanged grid or node set can keep its order through it. On the node sets tested it can. The seeds are the $t = 0$ profiles of polynomial-in-time solutions, continued through the edge by ordinary differential equations rather than by Taylor expansion, and they reduce to the interface-aware stencils of a jump as $\delta \rightarrow 0$: one construction covers the jump, the sub-grid edge and the resolved edge. In one dimension the seed stencils are fourth order at every resolution with the errors of the jump case while FD4 has its knee at $h = \delta$. In two dimensions the seeded RBF-FD operator is stable at the standard hyperviscosity at every edge width, and through an edge the nodes never resolve the seeds sit at the seed operator’s own floor, $20\times$ below the standard scheme in the velocity and $67\times$ in the spurious tangential velocity at 19,600 nodes through a flat edge, and 3 to $12\times$ below it through a curved one. To our knowledge no earlier scheme builds a stencil basis for the time-domain wave equation through a sub-grid smooth edge by an ODE march while leaving the rest of the scheme standard; the ingredients (bases from local solutions, polynomial

Trefftz spaces, the harmonic mean) are each classical. On the node sets tested the rule is: seed when $\delta \leq h$.

The comparison is against coefficient sampling on the same nodes and against our implementation of the coefficient treatments of seismic finite differences (cell means over one and two cells, band-limited coefficients, a widened edge) on the same schemes, one contrast and one test problem; their sources analyse and run them on staggered grids. On these grids and node sets no coefficient treatment reaches the seeds' order; the best of them is second order at a jump in one dimension, and the seeds lie 3 to 700 times below it there and 3 to 6 times below it on the scattered nodes.

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